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SURVEY

SoK: Advances in Anomaly Detection Techniques for Cryptoasset Transactions

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ABSTRACT Cryptoasset networks now settle hundreds of billions of dollars each day and underpin a rapidly expanding DeFi ecosystem. However, their openness exposes them to fraud, market manipulation, and protocol-level exploits. This Systematization of Knowledge (SoK) maps the state of anomaly detection in this environment. After outlining blockchain data characteristics and the full threat spectrum, we apply a reproducible OpenAlex search and multi-stage screening to collect 103 peer-reviewed studies. These works are organized into four methodological families: statistical analysis, network analysis, machine learning, and heuristic-based, which we compare across data assumptions, detection scope, interpretability, scalability, and robustness. Five cross-cutting gaps emerge: label scarcity, adversarial evasion, real-time scalability, behavioral ambiguity, and multi-chain visibility. We translate these gaps into a research agenda centered on hybrid graph-neural/heuristic pipelines, drift-aware statistics, explainable deep models, privacy-preserving analytics, and standardized benchmarks. This SoK provides both a concise snapshot of current techniques and offers perspectives on securing the next generation of blockchain infrastructure.

INDEX TERMS Anomaly, crypto-asset, graph theory, machine learning.

I. INTRODUCTION

A. BACKGROUND AND MOTIVATION

In 2008, blockchain technology was introduced by Satoshi Nakamoto as the foundational distributed ledger underpinning Bitcoin cryptoasset transactions [1]. This groundbreaking implementation enabled Bitcoin to address the longstanding double-spending problem, where the same digital asset could be spent more than once without relying on a trusted third-party authority or centralized intermediary [2], [3]. Blockchain technology achieves trustless verification through cryptographic techniques, decentralized consensus protocols (such as proof-of-work and proof-of-stake), and transparent yet pseudonymous transaction records stored across numerous network nodes. Due to these attributes, blockchain rapidly found applications outside of cryptoassets, finding widespread adoption across diverse fields including finance [4], [5], supply chain management [6], [7], [8], healthcare [9], [10], and decentralized applications (DApps) [11], [12].

Despite its adoption in various sectors, cryptoasset remains blockchain's most prominent and widely recognized application. Transaction networks, the graphical representation of transactions between blockchain addresses or entities, have emerged as critical analytical tools for understanding complex patterns and dynamics within cryptoasset ecosystems. These networks offer insights into economic activity, asset distribution, and user behavior patterns [13], [14], [15] at a granularity unattainable with traditional financial monitoring systems [16], [17], [18]. Moreover, analyzing these transaction networks helps reveal subtle structures and anomalies that may indicate suspicious or illicit behaviors, which traditional centralized monitoring mechanisms could overlook.

However, the intrinsic characteristics of blockchain systems, such as pseudonymity and decentralization, also create vulnerabilities exploitable by malicious actors. The cryptoasset ecosystem has grown significantly, briefly topping US$3

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trillion in total capitalization in late 2021 and now handling well over US$100 billion in daily on-chain value transfer. Within this large-scale environment, cryptoasset ecosystems have witnessed a notable rise in fraudulent activities, including money laundering, market manipulation, ransomware payments, and illicit financial transactions involving dark marketplaces and cybercrimes. Generally, in this context, an anomaly or anomalous transaction refers to activity exhibiting characteristics significantly divergent from what is deemed "normal," often indicative of aberrant behavior. Determining anomalous status can depend on contextual factors and specific transactional or market conditions. Financial transactions encompass various characteristics, with anomalies perceived differently depending on the metrics employed. A transaction flagged as anomalous under one set of criteria may not meet the same designation under another framework. Regulatory bodies worldwide are tasked with scrutinizing these anomalies within financial transactions and implementing requisite interventions.

Within cryptoassets, these anomalies are commonly categorized into three main types. Point anomalies are individual transactions markedly deviating from a typical profile, such as an unusually large single transfer or a transaction involving a previously inactive wallet. On the other hand, contextual anomalies appear anomalous primarily due to their context, like occurring at unusual times or representing sudden high-frequency activity from typically inactive accounts. Finally, collective anomalies include sequences or groups of transactions that seem suspicious when viewed together, even if individual transactions appear normal, such as coordinated pump-and-dump schemes or layering activities used in money laundering.

High-profile incidents involving cryptoasset exchanges and decentralized finance (DeFi) platforms, such as the Mt. Gox Collapse (2014) [19] and the hacks of Poly Network (2021) [20], serve as stark examples of these vulnerabilities and the resulting anomalies. Such events have resulted in losses totaling billions of dollars, undermining trust and illustrating significant weaknesses in existing anomaly detection frameworks. Consequently, a growing urgency and importance is placed on developing robust anomaly detection systems tailored explicitly for blockchain transaction networks.

Consequently, a growing urgency and importance is placed on developing robust anomaly detection systems tailored explicitly for blockchain transaction networks. Effective anomaly detection systems help safeguard users and businesses by detecting illicit activities in near real-time, thereby maintaining market integrity, enhancing regulatory compliance, and bolstering overall ecosystem security. The increased complexity, rapid evolution, and immense transaction volume within blockchain systems necessitate innovative detection methodologies. Researchers have thus explored various techniques ranging from classical statistical analysis and heuristic-based rules to more sophisticated machine learning and deep-learning approaches, including hybrid models for financial trend prediction [21], as well as network-theoretical analyses explicitly designed for graph-structured blockchain data [18].

Nevertheless, despite considerable efforts in developing such techniques, existing literature remains fragmented and lacks a unified synthesis of knowledge. Different methods are evaluated under varying assumptions, datasets, and experimental conditions, making direct comparisons difficult and highlighting the need for a systematic review. This study addresses precisely this gap by comprehensively reviewing existing literature on anomaly detection within blockchain transaction networks. By systematically classifying and analyzing existing methods, identifying limitations in current research, and highlighting open research challenges, we aim to provide clear guidance for future research directions, potentially aiding future work on more effective, scalable, and interpretable anomaly detection systems for blockchain ecosystems.

B. SCOPE AND CONTRIBUTION

This Systematization of Knowledge (SoK) provides a comprehensive review and analysis of anomaly detection techniques specifically targeting cryptoasset transaction networks. Our scope centers on analyzing transaction data such as graphs and time series to identify illicit activities, network attacks, or protocol misuse, primarily within prominent cryptoassets like Bitcoin and Ethereum, while also considering techniques applicable to other platforms. We deliberately exclude studies focused solely on market price prediction without transaction-level analysis or broader blockchain applications outside the financial/transactional domain, such as supply chain management. The key contributions of this work are:

- A systematic literature review identifying and synthesizing critical publications in blockchain anomaly detection, utilizing a rigorous, reproducible paper selection process, ensuring comprehensive coverage and reliability.
- A detailed taxonomy of anomaly detection techniques employed within blockchain transaction networks, clearly articulating methodological distinctions and application contexts.
- A critical analysis highlighting key research gaps, methodological limitations, and emerging challenges faced by existing studies in this domain.
- Recommendations for future research directions that emphasize developing innovative approaches to address identified limitations, improve detection effectiveness, scalability, interpretability, and adaptability to diverse cryptoasset platforms, and integration with new technologies.

This systematic review aims to guide researchers, practitioners, and policymakers by clarifying state-of-the-art in cryptoasset anomaly detection and highlighting key areas for future investigation.

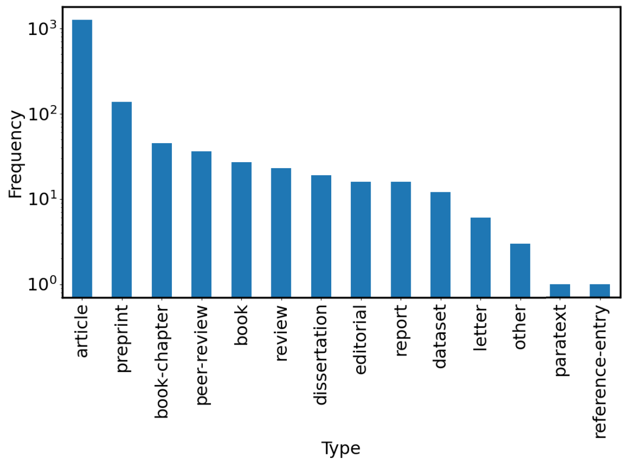

FIGURE 1. Distribution of document types among the 1,933 publications retained after preliminary screening. The vertical axis is shown on a logarithmic scale.

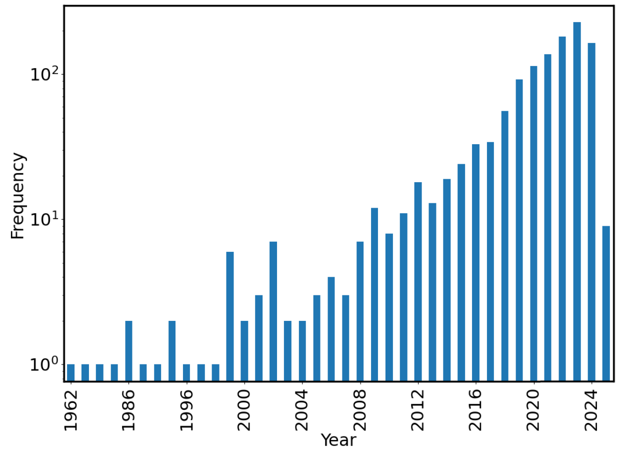

FIGURE 2. Distribution of the 1,438 selected research papers by publication year.

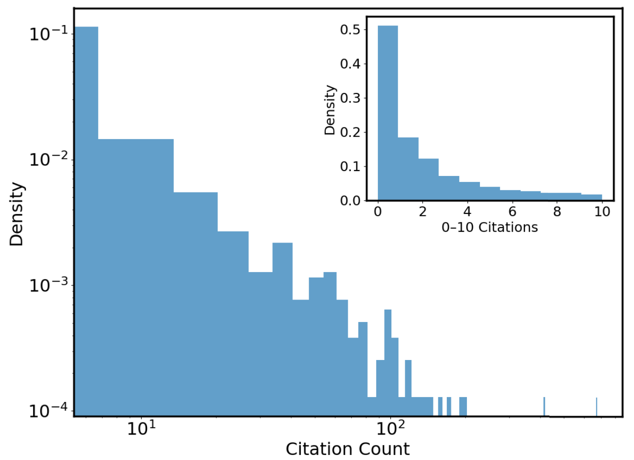

FIGURE 3. Citation count distribution of the 1,438 selected research papers, The inset provides a closer look at the 0–10 citation range.

C. METHODOLOGY

1) PAPER SELECTION PROCESS

To address our research questions on anomaly detection within the cryptoasset domain, we conducted a systematic literature search using the OpenAlex database. Our search string was designed to capture the full breadth of research at the intersection of anomaly detection, cryptoassets, and graph analytics. Specifically, we queried OpenAlex with:

```
(''anomaly detection'' OR anomaly OR anomalies OR
''detection of anomalies'' OR ``fraud detection''
OR forensics OR fraud OR ``money laundering'' OR
''market manipulation'' OR ``transaction network'')
AND (cryptocurrency OR crypto OR ``crypto asset''
OR ``crypto wallet'' OR bitcoin OR ethereum OR XRP
OR Solana OR Tether) AND (graph OR graphs OR
``graph based'' OR ``graph-based'' OR
networks OR network)
```

This search returned a total of 5,020 publications (as of as of March 6, 2025). We then applied a multi-stage screening process as follows. First, we excluded publications lacking a title, author, abstract, DOI, or indexing, as well as any duplicates (FC1). Next, we excluded all non-English publications (FC2). These preliminary filters were implemented to ensure that the final selection included only records with complete and accurate information suitable for further analysis. After these steps, 1,933 publications were retained, spanning various document types, as shown in Fig. 1.

Next, we selected publications categorized as articles, preprints, or book chapters (FC3). Book chapters were included because many conference papers are published in this format. After narrowing these categories, we excluded review or survey papers (FC4), resulting in 1,438 research papers, and 215 review/survey papers were removed. By plotting the publication years of these papers as shown in Fig.2, we observe a notable growth starting in 2009, the year Bitcoin was introduced. Note that the apparent drop for 2025 is due to the partial data collected early in that year and does not reflect a decline in research interest. Consequently, we excluded papers published prior to 2009 (FC5). Finally, we applied a minimum citation threshold of three (FC6), based on the citation distribution illustrated in Fig.3, which yielded a refined set of 509 publications. Finally, from this pool of papers, we applied the following criteria to ensure relevance to our study: a primary focus on blockchain transaction network analysis, a clearly defined methodology for anomaly detection, and either an empirical evaluation or a theoretical foundation (FC7). As a result, we obtained a final set of 103 publications on anomaly detection in cryptoassets. These publications were examined and compared based on their methodology, data sources, and reported performance. The complete selection process is illustrated in Fig.4.

2) CLASSIFICATION FRAMEWORK

We developed a multi-dimensional classification framework to systematically organize and analyze the diverse body of research on anomaly detection in cryptoasset transaction networks. This framework distinguishes between primary dimensions, which represent fundamental characteristics

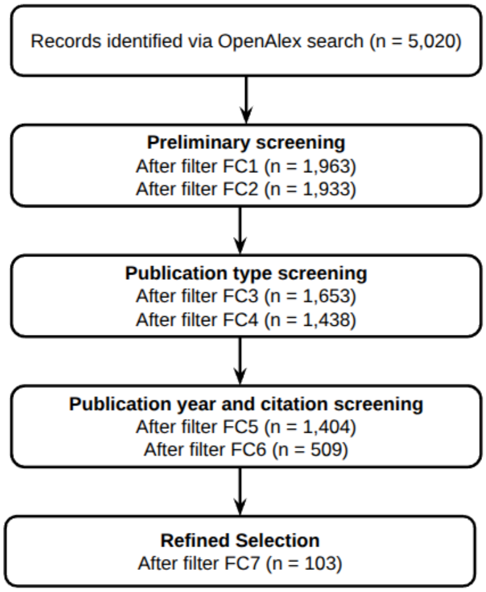

FIGURE 4. Literature review workflow.

dictating the core nature of the detection approach, and secondary dimensions, which offer complementary perspectives for finer-grained analysis. The primary dimensions guiding our classification are:

- **Detection Methodology:** This is the cornerstone of our classification and forms the primary axis for the detailed review presented in Section III. We categorize techniques based on their core algorithmic approach into statistical methods, network analysis techniques, machine learning approaches (including supervised, unsupervised, and deep learning), and heuristic-based strategies. We consider methodology paramount because it fundamentally shapes the detection process, dictating data requirements, computational complexity, interpretability, and the types of anomalies a technique is best suited to identify. It reflects the core technical innovations and provides a clear structure for comparing research contributions.
- **Data Sources:** Another critical primary dimension distinguishes whether methods rely on on-chain data (publicly available on the blockchain ledger), off-chain data (external sources like market prices, social media, or proprietary information), or a hybrid combination. The data source fundamentally limits or enables the scope of detectable anomalies.
- **Application Domain:** This dimension classifies studies based on their intended target area, directly relating to the types of anomalies (as defined in section I:A) they aim to detect. For instance, techniques focused on financial fraud detection might search for specific point anomalies in transactions or particular patterns of collective anomalies suggestive of illicit fund flows. Studies targeting market manipulation analysis often seek particular collective or contextual anomalies indicative of artificially influenced market activity. Furthermore, the domain of network security enhancement typically involves detecting contextual or collective anomalies related to protocol misuse or network-level attacks. Classifying studies by their application domain clarifies the practical objective of the proposed techniques and the specific kinds of divergent or aberrant behaviors they are designed to identify within the blockchain ecosystem.

While the primary dimensions, particularly methodology, provide the main structure for this SoK, analyzing the literature through secondary dimensions offers valuable additional insights and reveals further nuances:

- **Temporal Aspects:** Studies can be viewed based on whether they perform a static analysis on a snapshot of the network or employ dynamic analysis to capture temporal evolution and behavioral changes over time.
- **Scale of Analysis:** Techniques may operate at different granularities, focusing on node-level behavior (individual addresses/transactions), subgraph patterns (local neighborhoods or communities), or network-wide properties.

Nonetheless, while recognizing the value of these multiple perspectives, this SoK adopts the detection methodology as the central organizing principle for the in-depth discussion as it allows for a focused comparison of the core technical advancements and sets a clear direction for evaluating the state-of-the-art in cryptoasset anomaly detection.

II. DEFINING AND CHARACTERIZING ANOMALIES IN BLOCKCHAIN TRANSACTION NETWORKS

A. BLOCKCHAIN AND CRYPTOASSET FUNDAMENTALS

A blockchain is a type of Distributed Ledger Technology (DLT) that records transactions in a decentralized and immutable manner. Security is maintained through a chain of cryptographically linked blocks, where each block contains transaction data, a timestamp, and a hash of the preceding block. This structure makes tampering with historical data computationally infeasible.

Blockchain systems primarily use two transaction models, as illustrated in Fig.5.

- **Unspent Transaction Output (UTXO) model:** Used by Bitcoin, this model tracks discrete chunks of cryptoassets. Each transaction consumes existing UTXOs and generates new ones, providing clear asset traceability which is valuable for forensic analysis.
- **Account-based model:** Used by Ethereum, this model functions like a bank account, maintaining a balance that is directly debited or credited. This approach simplifies state management, especially for applications involving smart contracts.

Network integrity and agreement on the ledger's state are ensured by consensus mechanisms. The two most common are **Proof-of-Work (PoW)**, which relies on computational power (mining) to validate transactions, and **Proof-of-Stake**

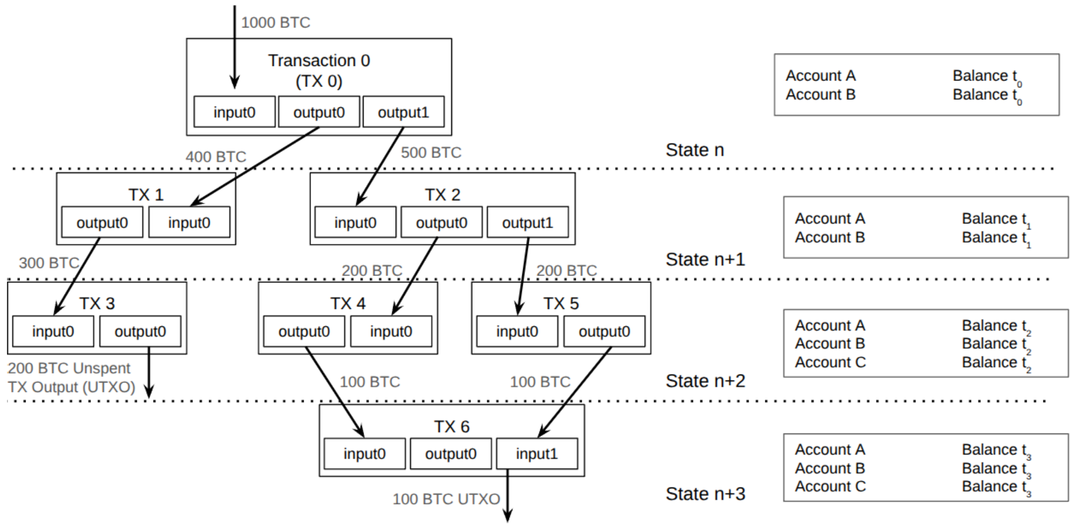

FIGURE 5. Illustration of the Unspent Transaction Output (UTXO) model (left) and the account-based model (right). On the left, each transaction consumes specific outputs (e.g., 400 BTC, 500 BTC) and creates new outputs, some of which remain unspent. On the right, the system tracks account balances, transitioning from one global state (State n+1) to the next (State n+2, n+3) as transactions occur.

(PoS), where participants stake their own cryptoassets to secure the network. Both are designed to prevent fraudulent activities like double-spending. PoS generally consumes significantly less energy than PoW and can potentially offer better scalability. However, it also introduces different security considerations and potential risks, such as the "nothing at stake" problem, though mechanisms exist to mitigate this.

Transactions on blockchains are pseudonymous, not anonymous. Users operate via cryptographic addresses, which are not directly tied to real-world identities. However, patterns in transaction data can be analyzed to cluster addresses and potentially de-anonymize users, a key consideration for forensic investigation. Furthermore, platforms like Ethereum support two distinct account types: **Externally Owned Accounts (EOAs)**, which are controlled by users via private keys, and **Smart Contract Accounts**, which are governed by their own embedded code. Smart contracts are self-executing programs that enable decentralized applications (dApps) by automating complex logic. A key distinction relevant to anomaly detection is that only EOAs can initiate transactions; smart contracts can only react to transactions they receive from EOAs or other contracts. This interaction creates unique on-chain patterns and introduces vulnerabilities that can be exploited, making smart contract behavior a significant source of detectable anomalies. For a more comprehensive overview of blockchain technology, its architecture, and diverse applications, we refer the interested reader to foundational reviews [22], [23].

B. COMMON ANOMALIES/ATTACKS

Classifying anomalies into point, contextual, and collective provides a valuable framework for understanding how malicious behaviors may manifest in blockchain networks. However, in practice, many anomalies straddle multiple categories and target different ecosystem levels, from individual transactions to consensus mechanisms and smart contracts. The following part will discuss common anomaly and attack types, illustrating how they map to the anomaly categories and how they manifest in real-world scenarios.

1) TRANSACTION-LEVEL FRAUD AND ABUSES

- **Double-Spending Attempts (Point/Contextual):** This type of anomaly involves an attacker broadcasting two conflicting transactions that spend the same coins, aiming to invalidate one transaction after a recipient believes it is confirmed. Although consensus mechanisms like PoW or PoS are designed to mitigate such attacks, low confirmation times or chain reorganizations may allow double-spending to succeed. In practice, this anomaly often appears as nearly identical transactions issued in rapid succession from the same address, frequently with one transaction replaced by another (e.g., via higher fees). Successful double spending can lead to direct financial losses for merchants or service providers that accept unconfirmed transactions.
- **Single Large/Outlier Transfers (Point):** Single large transfers that greatly exceed an address's historical transaction sizes often represent point anomalies. They are particularly suspicious when coming from an address known for relatively modest activity or when the receiving address is newly created or previously inactive. Such outlier transactions may indicate an exchange hack, insider trading, or liquidation of illicitly obtained funds. A well-known historical example is the Mt. Gox hack, where enormous amounts of Bitcoin were siphoned from the exchange's hot wallets over time. Although some of the transfers were not obviously

suspicious at first glance, subsequent investigations revealed a pattern of repeated large outflows that ultimately contributed to the collapse of the platform. Because a single outlier can trigger heightened scrutiny, attackers sometimes break large amounts into smaller, timed movements to evade detection—highlighting how anomalies can evolve into more complex patterns when attackers act repeatedly.

- **Phishing/Dusting Attacks (Point):** Phishing in the cryptoasset context can involve unsolicited messages prompting users to send funds or sign malicious transactions while dusting attacks entail sending trivial "dust" amounts of cryptoasset to numerous addresses. Though each dust transaction is small, they can reveal wallet ownership links collectively if recipients consolidate dust in a single output. These attacks often coincide with unusual traffic spikes of micro-transactions to unconnected addresses, marking a contextual anomaly when viewed against normal transaction profiles. The initial dust might seem innocuous; however, combining these minute inputs can help adversaries de-anonymize users, eventually setting the stage for larger attacks.

2) ILLICIT FINANCIAL ACTIVITIES

- **Money-Laundering (Collective):** Money laundering in cryptoassets frequently takes the form of layering, where funds are passed through multiple addresses or mixing services to obfuscate their origins. Individual transactions in a laundering scheme may appear unremarkable, but collectively, they show repeated splitting, merging, or identical sums moving in rapid succession. These multi-hop, near-simultaneous transfers suggest that a cluster of addresses is cooperating to conceal the trail. Although on-chain mixers can add further complexity, certain transaction-flow signatures, like uniform amounts or synchronized timing, help forensic analysts flag these collective anomalies.
- **Terrorist Financing and Dark-Market Payments (Contextual/Collective):** Illicit financing for extremist groups or dark-market purchases often entails contextual anomalies, wherein funds move to or from addresses known for high-risk activity around specific events. Isolated transactions might appear normal, but a closer look at timing, such as spikes in donations during notable extremist events, can confirm suspicious intent. In many cases, intelligence from external sources (e.g., law-enforcement watchlists, dark-web scraping) reveals links that are not evident from on-chain data alone. Thus, these anomalies often require merging blockchain analysis with off-chain intelligence to achieve reliable detection.
- **Ransomware Payments (Point/Contextual):** When attackers encrypt victims' data and demand cryptoasset in return for decryption keys, the resulting ransomware payments typically present as large, one-off transactions to an address tied to a known strain of ransomware. Although each payment might be a point anomaly, viewing them collectively can also reveal patterns. For instance, the same address receives repeated payments from geographically dispersed victims. This dual perspective underscores how many anomalies cross the line between point and collective categories, especially if attackers systematically reuse addresses or quickly launder collected funds.

3) MARKET MANIPULATIONS (MM)

- **Pump-and-Dump Schemes (Collective):** Pump-and-dump schemes rely on a coordinated group rapidly buying an illiquid token, driving up the price (the pump), then selling off their holdings en masse (the dump). While each purchase or sale alone could resemble normal trading activity, the collective effect is abrupt price and volume spikes followed by a dramatic crash. These schemes often involve off-chain coordination on social media or messaging platforms, combining an on-chain collective anomaly with an external organizational layer [24]. Exchanges or regulators monitoring trade volume patterns and market sentiment can sometimes detect these schemes early, although many happen too quickly for timely intervention.
- **Wash Trading (Contextual/Collective):** Wash trading involves the same party (or colluding parties) repeatedly buying and selling an asset to inflate volume or stabilize prices. Although each transaction can look typical, the combined pattern reveals frequent trades between the same addresses with minimal net movement of funds. This is especially common in newer token markets or NFT marketplaces wanting to project artificial liquidity. If a blockchain records all trades transparently, analyzing repeated address pairs, cyclical flows, or near-zero net gains can expose the manipulative nature of wash trading.
- **Front-Running and MEV (Contextual):** Front-running arises when an entity (often a miner, validator, or specialized bot) reorders transactions in a block to exploit opportunities such as large swaps on a decentralized exchange. These reorderings create small time windows where an attacker can insert a transaction that profits from price movements [25]. Observing repeated occurrences of newly inserted transactions just before large user swaps indicates a contextual anomaly, which is normal from a transactional standpoint but suspicious in block order. Miner/validator extractable value (MEV) can become systemic if left unchecked, impacting DeFi markets by consistently disadvantaging ordinary users.

4) NETWORK/CONSENSUS ATTACKS

- **51% Attacks (Collective):** Attacks against the consensus layer, such as 51% attacks or selfish mining, are not strictly transactional anomalies but significantly affect transactions when malicious miners rewrite or withhold blocks. A sudden concentration of hashing or staking power can lead to chain reorganizations, invalidating previously confirmed transactions or enabling double-spending. In the case of selfish mining, an entity mines blocks in private and selectively publishes them to the network to gain disproportionate rewards. Monitoring block production and correlating it with unusual transaction reversals can expose these anomalies, which often involve both protocol-level irregularities and suspicious transaction patterns.

- **Selfish Mining or Block Withholding (Contextual):** In selfish mining, a miner (or pool) withholds newly mined blocks from the public network, secretly building a private branch of the chain. By selectively releasing these blocks later, the miner can create reorganizations that invalidate others' blocks and claim more rewards. Block withholding shares similar dynamics; miners choose not to publish certain blocks, potentially to collude or manipulate difficulty. These behaviors represent contextual anomalies because they deviate from the expected block-discovery pattern; a single withheld block might seem benign, but repeated withheld or privately mined blocks can yield persistent advantages. Over time, such strategies threaten network fairness and reduce security assurances for honest participants.

5) SMART-CONTRACT VULNERABILITIES

- **Reentrancy Attacks (Contextual/Collective):** Reentrancy vulnerabilities arise when a contract sends funds (or triggers external calls) before updating its state. Attackers can exploit this to repeatedly call the same function (e.g., a withdrawal method) within a single transaction or block, draining the contract's balance in small increments. Although each call may look like a normal contract interaction, the sequence viewed collectively reveals an anomalous pattern of repetitive withdrawals in a very short timeframe. A reentrancy exploit may begin as a single suspicious call (contextual anomaly) but often escalates into a chain of exploit calls that constitute a collective anomaly.
- **Integer Overflow/Underflow (Point/Contextual):** Poorly coded smart contracts that do not enforce safe arithmetic can allow counters or balances to "wrap around" when they exceed a maximum integer value (overflow) or drop below zero (underflow). Such sudden, erratic changes in contract state variables—like a token balance jumping from near-zero to a huge number—can stand out as a point anomaly. If multiple overflow exploits occur rapidly, they may also be viewed as contextual anomalies. These attacks can quickly cripple a contract's logic, enabling unauthorized minting of tokens or erroneous payouts.
- **Ponzi and Pyramid Schemes (Collective):** Certain decentralized applications promise outsized returns to early participants at the expense of later ones, effectively functioning as Ponzi or pyramid schemes. Analyzing on-chain flows reveals a systematic pattern in which initial investors receive payouts drawn entirely from the capital contributed by newer participants. Each individual deposit may not look out of place, but collectively, the scheme exhibits unsustainable "rewards" with minimal legitimate revenue. Detecting these vulnerabilities involves tracking net inflows and outflows over time, often requiring address clustering to identify repeated participants.

TABLE 1. Anomalies listed in public tagpacks.

Anomaly	# Case	# Addresses
Gambling	12	2,659
Mixing service	7	36,763
Hack / theft	4	659
Scam	3	3,132
Sanction	3	599
Ransomware	3	14,653
Extremism	2	519
Sextortion	2	71,127
Fraud	1	6,699
Dark-web market	1	117
Pyramid scheme	1	8
Ponzi scheme	1	52
Phishing	1	6
Terrorism	1	155

6) ADDRESS CLUSTERING

Even though address clustering is not itself a direct attack, it can have significant security and privacy implications. By grouping multiple addresses likely controlled by the same entity, often identified through heuristics such as co-spending, shared key usage, or overlapping transaction patterns, investigators and adversaries can deanonymize users who believed they were pseudonymous. Moreover, studies of transaction networks frequently reveal centralizing tendencies despite the underlying blockchain's decentralized architecture: influential "hub" addresses (e.g., exchanges, mixers, or large custodial services) interact with a disproportionately high number of other addresses, effectively concentrating transaction flow. This hub-and-spoke structure diminishes the ideal of evenly distributed control, creating single points of failure or heightened regulatory focus. From a detection standpoint, clustering can pinpoint suspiciously large hubs or flows of illicit funds, yet from a privacy perspective, it can expose user relationships and compromise anonymity, underscoring how the same network analytics can be both a valuable investigative tool and a serious privacy concern.

These anomalies often combine or evolve across multiple categories, creating what can be termed "layered complexities." For instance, a single suspicious transaction may lead investigators to uncover a larger laundering operation or a 51% attack can be accompanied by deliberate double-spending attempts. Similarly, the boundaries between point, contextual, and collective anomalies can blur: while a

TABLE 2. Top 10 cryptoassets involved in anomalies listed in table 1.

Cryptoasset	# Case
Bitcoin (BTC)	25
Ethereum (ETH)	19
Litecoin (LTC)	5
Bitcoin Cash (BCH)	3
Monero (XMR)	2
Zcash (ZEC)	2
Bitcoin SV (BSV)	1
Bitcoin Gold (BTG)	1
Dash (DASH)	1
TRON (TRX)	1

reentrancy exploit may first appear as a single abnormal transaction call, repeated invocations reveal a collective pattern. Moreover, many incidents underscore the critical role of off-chain intelligence—such as social-media chatter or market announcements—in boosting detection accuracy for manipulative behaviors like pump-and-dump or wash trading. Consequently, effective monitoring requires combining on-chain analytics with external context to capture the full spectrum of illicit activities.

To complement/show the concept of the anomalies discussed above with the real-world example, we draw on the GraphSense TagPack [26], an open-source, community-maintained collection of machine-readable attribution tags. Each tag links one or more blockchain addresses to a real-world actor, e.g., an exchange, darknet market, or sanctioned entity.

Table 1 shows that sextortion and mixing-service activity dominate in terms of raw address counts despite appearing in only two and seven cases, respectively. Both anomaly schemes naturally generate long address chains (victims in sextortion campaigns, deposit/withdrawal wallets in mixers), inflating their footprint relative to more concentrated events such as exchange hacks. Conversely, only a handful of addresses represent categories like pyramid or phishing, illustrating the long-tail of niche but still security-critical threats.

Turning to platform distribution, Table 2 shows that Bitcoin and Ethereum are the main platforms for the anomalies. Together, they account for 44 of the 61 annotated cases (72%), making them the primary proving ground for detection research. The presence of privacy-enhancing chains (Monero, Zcash) with at least two packs each highlights the growing forensic interest in assets explicitly designed to obfuscate flows. The cases listed here highlight the breadth of today's threat landscape and foreshadow the twin challenges of scalability and cross-ledger generalization that motivate the taxonomy for how anomaly detection methods are organized and presented next.

C. TAXONOMY OF BLOCKCHAIN TRANSACTION ANOMALY DETECTION TECHNIQUE

To systematically analyze anomaly detection methods applied to blockchain transaction networks, we propose a comprehensive taxonomy depicted in Fig.6. This taxonomy categorizes existing anomaly detection techniques into four primary groups based on their core methodological approaches: *statistical analysis*, *network analysis*, *machine learning*, and *heuristic-based* methods. Each primary category further comprises subcategories that reflect specific methodological characteristics, analytical strategies, or underlying theoretical principles.

Our rationale for selecting these categories is based on methodological clarity and practical relevance observed in the existing literature. Statistical methods employ quantitative analysis, anomaly scoring, and probabilistic modeling to identify deviations from expected transaction patterns. Network analysis techniques leverage blockchain transaction graphs' structural and topological characteristics to identify anomalous entities or interactions. Machine learning methods encompass data-driven algorithms that autonomously learn patterns from historical transaction data, including supervised, unsupervised, and deep learning frameworks. Finally, heuristic-based approaches utilize rule-based or expert-defined models, often integrating analytical models or cryptographic properties intrinsic to specific blockchain platforms.

The frequency of research publications across these categories, shown in Fig.7, indicates clear trends and research priorities within the cryptoasset anomaly detection literature. Machine learning methods dominate, accounting for 49 out of 103 analyzed studies, reflecting the increasing emphasis on adaptive, data-driven techniques capable of handling large-scale, complex transaction data. Network analysis constitutes the second-largest group with 30 studies, emphasizing the importance of graph-based perspectives and the structural analysis of blockchain transaction relationships. Heuristic-based approaches, 14 papers, and statistical techniques, 10 studies, while fewer in number, still provide significant insights, particularly in contexts requiring transparency, interpretability, or well-defined probabilistic frameworks.

Each primary category is subdivided to reflect specific methodological details. Within machine learning approaches, supervised learning methods rely on labeled training data, making them effective but data-intensive. Unsupervised and semi-supervised learning approaches address data scarcity by identifying intrinsic patterns or anomalies without extensive labeled data. Network analysis methods focus on varying analytical scales (node-level, subgraph-level, and network-wide) and consider both static snapshots and dynamic, evolving blockchain environments. Heuristic-based methods employ analytical rules derived from expert knowledge or cryptographic principles, offering transparency and interpretability. At the same time, statistical approaches use rigorous mathematical frameworks such as distribution-based anomaly detection, time-series forecasting, and statistical profiling to quantify transaction anomalies systematically.

While this structured taxonomy aids in clearly understanding and organizing existing methods, overlaps and hybridization among categories exist. For instance, graph

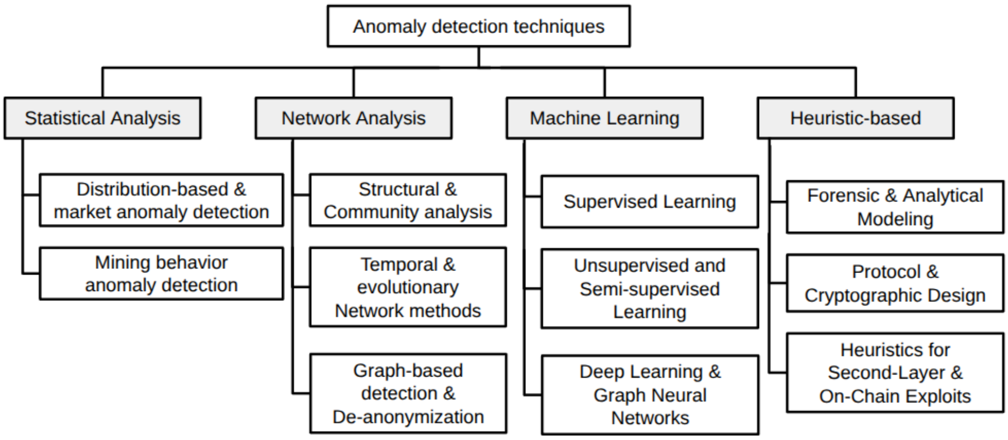

FIGURE 6. **Taxonomy of anomaly detection techniques.**

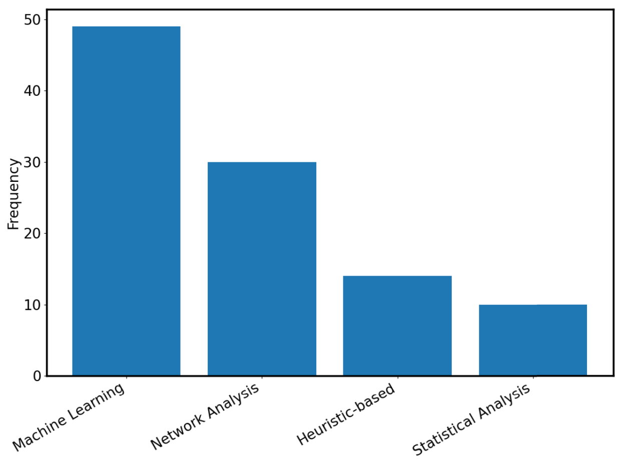

FIGURE 7. **Distribution of the 103 selected research papers across the categories based on the proposed taxonomy.**

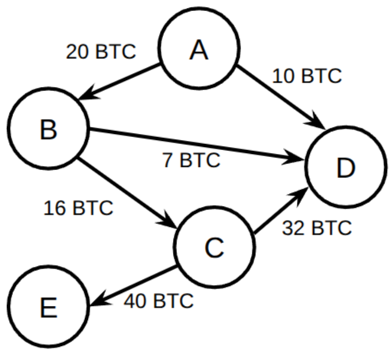

FIGURE 8. **Example of a directed transaction graph among five addresses (A, B, C, D, E). The edges are weighted by the amount of BTC transferred from one address to another.**

neural networks combine machine learning and network-analytic techniques, indicating evolving interdisciplinary approaches. Such intersections underscore the dynamic nature of the field, revealing opportunities for future methodological innovation. The subsequent section systematically explores each category in-depth, comparing methodologies, highlighting their strengths and limitations, and identifying key research gaps.

D. CONCEPTS IN STATISTICAL METHODS

Statistical analysis approaches in cryptoasset networks commonly center on modeling typical transaction or market behaviors via probability distributions, time-series analyses, or multivariate control frameworks. By measuring deviations from these established "normal" baselines—whether through simple metrics like mean and variance or more advanced techniques like autoregressive models and tensor-based analyses—these methods can highlight anomalous spikes or patterns that suggest fraudulent activity. Crucially, statistical approaches serve as one of the earliest lines of defense and can be adapted to flag both sudden outliers (point anomalies) and more nuanced deviations that unfold over time.

E. TRANSACTION GRAPHS AND NETWORK CONCEPTS

Analyzing cryptoasset transactions through the lens of network science provides powerful tools for understanding the flow of value, identifying influential participants, and detecting anomalous activities. By representing blockchain data as graphs, we can leverage well-established graph theory concepts and algorithms to gain insights that might be hidden when examining individual transactions in isolation.

Cryptoasset transaction networks can be naturally modeled as graphs. Typically, such graphs are constructed by aggregating transactions occurring within a specific time window, e.g., hourly, daily, or weekly, to create a snapshot of network activity. Mathematically, this snapshot is represented as $G = (V, E)$ where nodes V represent addresses or transactions and edges E represent relationships or interactions, such as the transfer of funds. For example, if user A sends x amount of tokens to user B, to user B, we represent this as a directed edge from node A to node B weighted by x, see Fig.8 for the simple illustration of the transaction graph. Edges in cryptoasset graphs are usually directed due to the nature of transactions, i.e. sender to receiver, and often weighted by transaction value or frequency. This directed nature allows tracking the flow of funds clearly from origin to destination.

Several standard graph representations are employed to analyze blockchain data:

- **Transaction Graph:** Nodes represent individual transactions, and edges indicate the flow of funds between transactions.
- **User Graphs:** Nodes represent blockchain addresses or users, and edges reflect interactions or transfers between addresses/users.
- **Interaction Graphs:** Abstract representation where nodes can represent entities such as wallets, exchanges, or contracts, with edges denoting various forms of interactions.

These graphs are constructed by parsing transaction data recorded on the blockchain ledger. Each transaction typically links one or more input addresses to one or more output addresses.

Several basic graph metrics help characterize transaction networks:

- **Degree (k):** Number of edges connected to a node. For directed graphs, one can distinguish between in-degree, the number of incoming edges, and out-degree, the number of outgoing edges. The degrees of node v are defined as:

$$k_{in}(v) = \sum_{u \in V} e_{uv}, k_{out}(v) = \sum_{u \in V} e_{vu} \tag{1}$$

where e_{uv} is the edge from node u to node v and vice versa. The degree of node v can be, then, defined as $k(v) = k_{in}(v) + k_{out}(v)$

- **Strength:** Extends degree by summing edge weights connected to a node, useful for capturing transaction volume.
- **Clustering Coefficient (C):** Measures how closely connected a node's neighbors are to each other, capturing the local density of interactions. The clustering coefficient of node v is defined as:

$$C(v) = \frac{2T(v)}{k(v)(k(v)-1)} \tag{2}$$

where $T(v)$ represents the number of triangles through node v.

Centrality measures help identify important nodes within transaction networks:

- **Degree Centrality:** Nodes with higher degree centrality actively participate in more transactions, highlighting potential hubs.

$$C_D(v) = \frac{k(v)}{N-1} \tag{3}$$

where N is the total number of nodes.

- **Closeness Centrality:** Indicates how close a node is to all other nodes, useful for identifying influential nodes or key intermediaries.

$$C_C(v) = \frac{N-1}{\sum_{u \in V} d(u, v)} \tag{4}$$

where $d(u, v)$ is the shortest path distance between nodes u and v.

- **Betweenness Centrality:** Measures the extent to which a node lies on paths connecting other nodes, pinpointing nodes critical for information or value flow.

$$C_B(v) = \sum_{s \neq v \neq t \in V} \frac{\sigma_{st}(v)}{\sigma_{st}} \tag{5}$$

where σ_{st} denotes the total number of shortest paths between nodes s and node t, and $\sigma_{st}(v)$ is the number of those paths passing through node v.

Community detection in cryptoasset networks involves identifying clusters of addresses that exhibit a higher density of interactions among themselves than with the rest of the network. By leveraging techniques such as modularity optimization, spectral clustering, or label propagation, analysts can uncover patterns that suggest coordinated behavior—be it legitimate operational clusters like exchanges or suspicious groups potentially involved in fraud or money laundering. This approach helps to simplify and elucidate the complex flow of transactions on the blockchain by highlighting both central hubs and isolated pockets within the network, thereby providing critical insights for regulatory compliance and risk management. Despite challenges like scalability and the inherently dynamic nature of blockchain data, community detection remains a powerful tool for discerning the underlying structure of transaction networks and enhancing the overall understanding of digital currency ecosystems. For the details on graph theory, refer to [27].

F. MACHINE LEARNING IN A NUTSHELL

Given the widespread adoption of machine learning for anomaly detection, a dedicated and detailed discussion is warranted. For a comprehensive theoretical background on machine-learning algorithms, readers are referred to the well-established textbooks [28], [29].

Machine learning has progressed rapidly over the past decade, finding applications at vastly different fields scales, from microscopic processes such as protein folding [30], [31], [32] and bacterial swimming behaviors [33], [34], [35], to human behavioral [36], [37], [38], drug delivery [39], [40] and onward to enterprise-scale optimizations like supply-chain logistics and manufacturing workflows [41], [42]. In the context of cryptoasset analysis, machine learning offers the ability to uncover subtle and adaptive patterns of fraudulent or malicious behavior by learning from vast amounts of transaction data, even as illicit tactics evolve.

- **Supervised Learning:** In supervised ML, models such as Random Forest or Support Vector Machines

(SVMs) are trained on labeled examples of normal versus anomalous transactions. By "learning" how known anomalies differ from legitimate activity, these models can then generalize to flag novel suspicious cases. However, constructing a reliable labeled dataset, especially in decentralized cryptoasset settings, can be challenging due to the scarcity of confirmed fraud labels and the possibility that malicious actors continuously change their strategies.
- **Unsupervised Learning:** Unsupervised approaches detect anomalies by modeling what "normal" data looks like, then measuring how strongly each new transaction deviates from this norm. Clustering techniques like k-Means or DBSCAN group transactions according to similarity, labeling data points in low-density regions or forming their own small clusters as outliers. Likewise, distance-based methods such as k-Nearest Neighbors measure each transaction's distance to its neighbors: points whose distances surpass typical thresholds are considered anomalies. Unsupervised methods are particularly appealing when labeled anomalies are scarce or non-existent.
- **Semi-Supervised Learning:** In many real-world blockchain use cases, only a small subset of transactions can be confidently labeled, e.g., a handful of confirmed scam addresses. Semi-supervised algorithms use this limited information to guide the detection process. A common tactic is to train a model primarily on normal data (e.g., one-class SVMs or autoencoders). Hence, the system learns normal behavior and flags anything sufficiently different as suspicious. This approach aligns well with cryptoasset ecosystems, where legitimate transactions vastly outnumber known fraudulent cases.
- **Deep Learning:** Neural networks often excel at capturing complex, high-dimensional relationships. Simple feed-forward networks and Convolutional Neural Networks (CNNs) can process time-series or transactional features. In contrast, Recurrent Neural Networks (RNNs) or Long Short-Term Memory (LSTM) networks handle sequential data such as address activity over time. A particularly relevant direction involves Graph Neural Networks (GNNs), which encode both node (address) attributes and topological information (who transacts with whom and how often). GNN-based models can uncover small, densely connected pockets potentially involved in money laundering or other collusive behaviors that might elude less graph-aware methods.

Effective ML-based anomaly detection critically depends on feature engineering. While raw transaction data such as addresses, timestamps, and transaction amounts provide a starting point, additional transformations often boost performance. Common feature types include:

- **Temporal Features:** Capturing time-based patterns like transaction frequency, value changes over time, or burstiness can reveal deviations from historical norms.
- **Behavioral Profiles:** Aggregating typical transaction sizes, counterparty interactions, or timing for an address helps identify uncharacteristic activity that might signal account takeover or illicit use.
- **External Signals:** Incorporating off-chain data, such as social media sentiment or market news, can be crucial, especially for detecting coordinated events like pump-and-dump schemes.

It is important to note the interplay between ML and Network Analysis (discussed conceptually in Section II-E). While our taxonomy presents them as distinct methodological families for clarity, insights from network analysis are often crucial inputs for ML models. Specifically, graph metrics such as node centrality, clustering coefficients, or community structure derived from the transaction graph frequently serve as powerful engineered features for ML algorithms. This synergy allows ML models to leverage the structural properties of the transaction network identified through network analysis techniques.

Beyond the learning paradigms, several algorithm classes are frequently applied:

- **Distance-Based (k-NN):** A straightforward yet effective method is k-Nearest Neighbors, where each transaction (represented by its feature vector) is evaluated against its k closest neighbors. Transactions with anomalously large distances are flagged. While simple to explain, k-NN can become computationally demanding in large-scale blockchain applications unless efficient indexing or approximate methods are employed.
- **Clustering Methods (k-Means, DBSCAN):** In k-Means, data points are partitioned into k clusters by minimizing the distance to each cluster's centroid. Transactions far from their nearest centroid or assigned to extremely small clusters can be considered anomalies. DBSCAN, in contrast, forms clusters based on density—points in sparsely populated regions are automatically deemed outliers. This density-driven approach can help reveal groups of addresses interacting in a suspiciously tight circle, unconnected to the rest of the network.
- **Tree-Based Models:**
  - **Isolation Forest:** isolates data points by randomly splitting features; anomalies tend to be split from the rest of the data more quickly, thus receiving higher anomaly scores.
  - **Random Forest:** typically a supervised classifier, can also provide outlier scores based on how consistently a transaction is classified compared to others. In labeled settings, such as a training set of flagged addresses, Random Forest can be used directly to classify transactions as normal or anomalous.
- **Neural Networks Approaches:**

-- **Feed-forward NNs:** learn to map input features, e.g., transaction size, frequency, node attributes, to a score or label indicating anomaly likelihood.
-- **Graph Neural Networks (GNNs):** such as Graph Convolutional Networks (GCNs) or Graph Attention Networks (GATs) capture the relational structure among addresses. By iteratively combining information from neighbors, GNNs detect anomalies that might appear only when viewed within the broader transaction subgraph.

G. EVALUATION METRICS FOR ANOMALY DETECTION

Anomaly detection in cryptoasset networks typically involves identifying rare, illicit, or otherwise suspicious transactions within a much larger pool of benign activity. This setting poses unique challenges for model evaluation, as standard metrics may not accurately reflect performance under high class imbalance. The following subsections discuss common metrics and highlight how curve-based analyses can provide deeper insights into a detector's effectiveness.

In most real-world datasets, anomalies constitute only a small fraction of total transactions. For instance, malicious addresses or fraudulent trades may make up far less than 1% of on-chain activity. Such class imbalance can undermine naive metrics—especially accuracy—by rewarding models that favor classifying the majority of instances as "normal." Consequently, an anomaly detector might achieve deceptively high accuracy while scarcely flagging any actual anomalies. This imbalance also complicates the training process: many machine-learning algorithms assume relatively balanced classes, and their performance or convergence can degrade when one class overwhelmingly dominates the other.

Confusion Matrix (TP, TN, FP, FN): A confusion matrix provides a granular look at a detector's outcomes. Here, True Positives (TP) are correctly identified anomalies, True Negatives (TN) are correctly identified normal transactions, False Positives (FP) are normal transactions misclassified as anomalies and False Negatives (FN) are missed anomalies. The following are matrices that are based on the confusion matrix.

- **Accuracy:**

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{6}$$

  Although accuracy is the most commonly reported metric, it can be misleading in imbalanced scenarios. If anomalies represent only 1% of transactions, a naive detector that flags everything as normal could achieve 99% accuracy, despite failing to detect any suspicious activity.
- **Precision:** indicates the proportion of flagged anomalies that are truly anomalous, highlighting how well a detector avoids raising false alarms.

$$Precition = \frac{TP}{TP + FP} \tag{7}$$

- **Recall:** indicates the proportion of flagged anomalies that are truly anomalous, highlighting how well a detector avoids raising false alarms.

$$Recall = \frac{TP}{TP + FN} \tag{8}$$

- **F1-Score:** indicates the proportion of flagged anomalies that are truly anomalous, highlighting how well a detector avoids raising false alarms.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{9}$$

The other widely used metrics for evaluating detection performance are curve-based, offering a more holistic view of how a classifier's performance changes under varying decision thresholds. A Receiver Operating Characteristic (ROC) curve plots the true positive rate (recall) against the false positive rate (1 - specificity), and the Area Under the ROC Curve (AUC-ROC) summarizes this overall trade-off. Values closer to 1.0 indicate better discrimination ability. However, when the number of negative (normal) instances vastly outweighs the positives (anomalies), the ROC curve can yield an overly optimistic picture.

To address this limitation, many anomaly-detection studies employ the Precision-Recall (PR) curve and calculate the Area Under the Precision-Recall Curve (AUC-PR). The PR curve directly focuses on precision and recall over various thresholds, making it more informative in heavily imbalanced contexts. Unlike the ROC curve, which plots all positive and negative samples equally, a PR curve highlights how well the detector maintains high precision (i.e., keeps false positives low) at different levels of recall (i.e., detects a large fraction of actual anomalies). In scenarios where anomalies are rare, a high AUC-PR typically provides a clearer picture of a model's practical effectiveness than a high AUC-ROC alone.

H. A PRIMER ON HEURISTIC-BASED APPROACHES

Heuristic-based methods rely on expert-defined rules or domain insights to pinpoint suspicious behavior in blockchain transactions. Rather than learning a model purely from data, these approaches encode known "red flags," such as unusually high-frequency transactions, dusting attempts, or repetitive output patterns indicative of mixers or tumblers. These rules often stem from forensic experience or analysis of known attack patterns. Because they draw on real-world observations, heuristics can be especially effective at catching known scams or protocol misuse in their early stages. They are generally highly interpretable, as the logic is explicit. However, their reliance on predefined patterns makes them inherently brittle; they typically struggle to detect novel or unforeseen attack vectors that deviate from known tactics. Furthermore, as adversaries evolve, these rule sets require continuous updating by experts to ensure they remain effective. Consequently, heuristics often complement data-driven approaches rather than replacing them, for example, by acting as an initial filter.

III. CASES OF ANOMALY DETECTION ANALYSIS

In this section, we comprehensively review existing anomaly detection cases applied to cryptoasset transaction networks, structured according to the previously proposed taxonomy shown in Fig.6. While we categorize existing literature into four broad classes, statistical analysis, network analysis, machine learning, and heuristic-based, there is often considerable overlap in practice. For instance, some studies grounded in statistical analysis may integrate machine learning classifiers to enhance outlier detection or employ heuristic rules to filter initial datasets. Conversely, purely heuristic-driven methods might incorporate network metrics (e.g., modularity, centrality) for improved anomaly spotting. Our taxonomy thus serves as a conceptual guide rather than a rigid classification, reflecting the multifaceted nature of anomaly detection in cryptoasset ecosystems. To clearly distinguish between local fraud detection and systemic security risks, these methodologies can be viewed through a layered lens, i.e. transaction-layer methods target individual value transfers, network-layer analyses expose structural clustering and flow patterns, and protocol-layer approaches scrutinize consensus integrity and smart contract vulnerabilities.

A. STATISTICAL ANALYSIS

We examine a set of studies employing various statistical analyses to detect anomalies in cryptoasset transaction networks. These approaches often rely on fundamental statistical metrics, such as mean, variance, correlation, higher-order moments, or more advanced time-series and regression-based modeling to characterize "normal" behavior and flag outliers. A summary of the studies covered in this category is presented in table 3 and 4.

1) DISTRIBUTION-BASED AND MARKET ANOMALY DETECTION

- **Signature:** One example of a statistical approach for outlier detection involves using signature to encoding time-series data into a collection of iterated integrals [43]. A truncated signature $S(\mathbf{X})$ of order n for a path $\mathbf{X}(t) \in \mathbb{R}^d$ where $\mathbf{X}(t) = (X^1(t), \ldots, X^d(t))$ records d features, e.g. price or volume, overtime $t \in [0, T]$ is defined as:
$$S^n(\mathbf{X}) = (1, S^1(\mathbf{X}), S^2(\mathbf{X}), \ldots, S^n(\mathbf{X})) \tag{10}$$
where
$$\begin{aligned} S^1(\mathbf{X}) &= \int_0^T d\mathbf{X}(t), \\ S^2(\mathbf{X}) &= \int_0^T \int_0^{t_1} d\mathbf{X}(t_1) \otimes d\mathbf{X}(t_2), \\ S^3(\mathbf{X}) &= \int_0^T \int_0^{t_1} \int_0^{t_2} d\mathbf{X}(t_1) \otimes d\mathbf{X}(t_2) \otimes d\mathbf{X}(t_3), \\ &\vdots \end{aligned} \tag{11}$$
where $\otimes$ denotes the tensor product. This operation combines the vector differentials, i.e. changes in features, at different time points to capture higher-order relationships and interaction effects within the time-series path $\mathbf{X}(t)$. Higher-order terms capture multi-scale temporal dependencies. For instance, $S^2(\mathbf{X})$ quantifies volatility interactions. To reduce computational complexity, the randomized signature is often employed:
$$R(\mathbf{X}) = \mathbf{A} \cdot S^n(\mathbf{X}) \tag{12}$$
where $\mathbf{A}$ is a random matrix with entries drawn from a specified distribution (e.g., Gaussian or Rademacher). These signatures signatures were applied to Bitcoin price-volume time series to detect pump-and-dump schemes characterized by abrupt price inflations followed by sharp declines. Empirical evaluation demonstrates the method's effectiveness, achieving high anomaly-detection performance up to 0.88 F1 score.
- **Benford's Law:** Another relevant approach relies on Benford's Law to detect fraudulent activities and unusual behaviors in cryptoasset transactions [44]. This law predicts that the leading digits of many naturally occurring numerical datasets follow a logarithmic distribution:
$$P(d) = log_{10}(1 + \frac{1}{d}) \tag{13}$$
where d ranges from 1 to 9. Deviations from this expected distribution often serve as indicators of manipulation or anomalies. Cryptoasset transaction data generally fit the conditions for Benford's Law, given their inherently wide numerical range. The study of major cryptoassets such as Ethereum and Bitcoin from 2009 to 2018 revealed that transaction values closely adhered to Benford's distribution based on Mean Absolute Deviation (MAD) thresholds, indicating largely unmanipulated behavior. By contrast, certain other cryptoassets, e.g., TENX, VERI, and DOGE, were identified as non-conforming to Benford's law, aligning with previously reported scandals and lawsuits.
- **Mahalanobis distances:** Robust anomaly scores have also been developed using Mahalanobis distances (MD) in cryptoasset market price data [45]. Mathematically, MD measures the distance of a data point r from the center of a distribution, accounting for covariance:
$$MD(r) = \sqrt{(r - \mu)^T \Sigma^{-1}(r - \mu)} \tag{14}$$
where $\mu \in \mathbb{R}^n$ is the mean vector and $\Sigma \in \mathbb{R}^{n\times n}$ is the covariance matrix of a random vector $r \in \mathbb{R}^n$. The anomaly score A is then defined as:
$$A(r) = \frac{MD(r)}{\sqrt{n}} \tag{15}$$
This score effectively identifies significant deviations in cryptoasset returns from typical behavior, effectively flagging unusual market movements as anomalies. For instance, it successfully flagged drastic price surges

during the metaverse boom in late 2021. Moreover, incorporating MD-based anomaly constraints into portfolio optimization reduced annual portfolio volatility from over 90% annually to the $40-50\%$ range, underscoring the potential of these methods for risk-sensitive investors.

- **Auto-Regressive Moving Average:** Furthermore, anomalies in Bitcoin price have also been studied using forecasting methods such as Seasonal Auto-Regressive Integrated Moving Average with Exogenous Factors (SARIMAX) [46]. By incorporating information gathered from social media, detecting manipulative practices, such as pump-and-dump schemes, becomes highly effective. These anomalies were especially prevalent during economic crises and periods of intense speculation, including the market turbulence observed during the COVID-19 pandemic. The social media sentiment input improved detection capabilities, though its contribution was modest during periods of intense manipulation. Overall, the combined forecasting and sentiment-analysis framework achieved an F1-score of up to 93%, demonstrating the strong synergy between market data and external sentiment signals [47].
- **Hidden Markov Multi-linear Tensor Models:** An alternative statistical monitoring framework employs Hidden Markov Multi-linear Tensor Models (HMTM) [48] and Multivariate Exponentially Weighted Moving Average (MEWMA) control charts [49], [50]. The goal of HMTM is to model the relationships in Bitcoin transaction networks that change over time but where the underlying state of the network (e.g., normal, suspicious) is hidden. HMTM builds upon the Multi-linear Tensor Model (MTM) [51] which model the probability of a transaction between node i and j at time t as:

$$P(y_{ijt} = 1|\mathbf{x}_{ijt}, \mathbf{u}_i, \mathbf{u}_j, \mathbf{v}_t) = \mathbf{x}_{ijt}\beta + < \mathbf{u}_i, \mathbf{v}_t, \mathbf{u}_j > + \varepsilon_{ijt} \tag{16}$$

where y_{ijt} indicates the presence 1 or absence 0 of a transaction, $\mathbf{x}_{ijt}$ a vector of covariates, i.e., known factors, that might influence the transaction, e.g., example transaction size, time of day, etc., β is the coefficient vector that quantifies the effect of the covariates, $\mathbf{u}_i$ and $\mathbf{u}_j$ represent latent vectors describing the position of nodes i and j in an underlying latent space, $\mathbf{v}_t$ captures the latent rules governing node interactions at time t and εijt is the error term assumed normally distributed around zero. The MTM assumes a static network. The HMTM adds a Hidden Markov Model (HMM) to account for the fact that the network can be in different unobserved states $\mathbf{B}_t = \mathbf{Y}_t - \Omega_t$ where $\mathbf{B}_t$ represents anomalous deviations, $\mathbf{Y}_t$ is the observed transaction adjacency matrix, and Ω_t is the expected structure based on latent variables under normal hidden states. The latent state $L_t = (\mathbf{u}_t, \mathbf{v}_t)$ describes the hidden dynamics guiding network connectivity over time. By monitoring deviations using Hotelling's T^2 statistic, significant anomalies in cryptoasset transaction behaviors are flagged. The method flags Bitcoin transactions between 2011 and 2013 that significantly deviate from typical historical patterns as potential anomalies align with the Mt. Gox leaked transactions [52].
- **Vector Autoregressive:** Vector Autoregressive (VAR) models have been employed to evaluate behavioral anomalies driven by external economic factors, such as gas price surges in Ethereum-based decentralized autonomous organizations (DAOs) [53]. In this context, the VAR framework captures how present values of multiple variables (e.g., gas prices and DAO activity) depend on their past values:

$$\mathbf{y}_t = v + A_1\mathbf{y}_{t-1} + A_2\mathbf{y}_{t-2} + \ldots + A_p\mathbf{y}_{t-p} + \mathbf{u}_t \tag{17}$$

where $\mathbf{y}_t = (r_t^{gas}, a_t)$ is a vector containing log-returns of the gas price and user activity at time t, v and $A_1, \ldots, A_p$ are coefficient matrices and u_t represents white noise. The VAR model enables the test for Granger causality between gas price changes and DAO activity while also capturing lagged effects and inter-dependencies between these variables over time. Analysis of 5,580 transactions from 7,825 users in 191 DAOs revealed a surprising result: despite significant gas price surges (up to 8500% increases in 2020), the model showed only minor statistical influence of gas price fluctuations on DAO user activity levels. This insensitivity contradicts typical market self-regulation expectations, where higher transaction costs would theoretically deter participation.
- **Adjusted volume:** Notable terrorist attacks can also be identified using an event-study approach based on

TABLE 3. Distribution-based & Market anomaly detection.

Paper	Blockchain	Data source	Anomaly	Measure
[43]	BTC	Off-chain (social network, BTC price and volume)	pump-and-dump	Precision: 0.93, Recall: 0.94, F1: 0.88
[44]	various currencies including BTC and ETH	On-chain (transaction)	fraudulent	MAD with thresholds
[45]	15 currencies including BTC and ETH	Off-chain (price)	price anomaly	N/A
[47]	BTC	Off-chain (price and social network)	pump-and-dump	F1: 0.93 (max), MCC: 0.31 (monthly)
[52]	BTC	On-chain (transaction)	fraudulent (Mt.Gox)	N/A
[53]	ETH-based DAO	On-chain (transaction)	user behaviour vs. gas price	Granger causality
[54]	BTC, XRP	On-chain (transaction) and Off-chain (Global Terrorism Database)	terrorist attacks	nonparametric rank test

mean-adjusted volume (AV) of user u at day t [54]:

$$AV_{u,t} = ln(V_{u,t}) - ln(\hat{V}_{u,t}) \tag{18}$$

which captures deviations between observed and expected user-level volumes. The average abnormal mean-adjusted volume (AAV) is then formed by summing these daily $AV_{u,t}$ for each user in the group and normalized by the total number of users, and the cumulative abnormal volume (CAV) expands that perspective across longer periods by aggregating AAV values around an event window:

$$AAV_t = \frac{1}{N_t}\sum_{u=1}^{N} AV_{u,t} \tag{19}$$

$$CAV_t = \sum_{t=1}^{T} AAV_t \tag{20}$$

Calculating mean CAV for the two-week intervals before $[t-15, t-1]$ and after $[t+1, t+15]$ any terrorist attack isolates sharp bursts of transactional activity consistent with short-term planning and execution patterns. When applied to Bitcoin blockchain transactions, categorized into groups like exchanges, dark markets, mixers, gambling platforms, and other services, CAV can reveal significant spikes in abnormal transaction volumes through mixers and unregulated exchanges in the weeks preceding major terrorist events. A focused case study on the Sri Lanka Easter bombing demonstrates the approach in action, detecting suspicious transfers by a single user with no plausible alternative explanation; backward traces link the wallet to other crimes, while forward traces reveal subsequent conversion to Ripple (XRP) and additional mixing via a high-value deposit wallet, underscoring the effectiveness of on-chain analysis in illuminating terrorist financing structures.

2) MINING BEHAVIOR ANOMALY DETECTION

Anomalous mining strategies can undermine the security guarantees of proof-of-work (PoW) systems by distorting reward distribution or enabling attacks such as double-spending. To address this, various statistical frameworks have been developed to detect non-compliant miner behavior.

- **Miner Sequence Bootstrapping:** One such approach is Miner Sequence Bootstrapping (MSB) [55], which models each miner's block-discovery event as a Bernoulli trial with success probability proportional to its hash-power share. Under normal conditions, the probability of a single miner discovering consecutive blocks in rapid succession should be relatively small unless its hash power is exceptionally large. Mathematically, let C_i^T denote the number of times miner i mines consecutive blocks over a given period T, and let S_i^T represent the output of a reshuffled (bootstrapped) sequence of block assignments in the t-th trial. The MSB index is then defined as:

$$MSB_i^T = \frac{C_i^T - \langle S_i^T \rangle}{\sigma\left[S_i^T\right]} \tag{21}$$

where $\langle S_i^T \rangle$ and $\sigma\left[S_i^T\right]$ denote the mean and standard deviation of the bootstrapped consecutive-block counts, respectively. An MSB value significantly greater than zero, often assessed via a p-value derived from the normal or empirical distribution of the bootstrap samples, indicates that miner i is an outlier, which may imply undisclosed strategic behaviors, such as delaying block publication. This methodology is also extended to detect mining cartels by measuring how often two miners i and j appear in succession.

$$MC_{ij}^T = \frac{C_{ij}^T - \langle S_{ij}^T \rangle}{\sigma\left[S_{ij}^T\right]} \tag{22}$$

where C_{ij}^T is the actual times that two consecutive blocks are first mined by miner i and then by miner j. Applying this framework to cryptoassets, including Bitcoin, Ethereum and Litecoin and Bitcoin Cashreveals the presence of anomalous miners in all four cryptoassets, with particularly notable clusters of outliers in Bitcoin Cash. Some of these miners remain unidentified (tagged as 'Unknown'), implying hidden pools or ad-hoc collusions. While anomalies are also observed in Litecoin and Bitcoin, the patterns there appear less concentrated than in Bitcoin Cash. The framework is then extended further to include the analysis of Monacoin, adopting a related statistical test based on a type II binomial model to detect disproportionate sequences of consecutive blocks [56]. A salient finding is that Monacoin exhibits the highest fraction of suspicious miners, corroborating the network's self-reported selfish mining incidents. Furthermore, many of these flagged entities exhibit collaborative structures, suggesting coordinated withholding of blocks among multiple miners.
- **Miner Share Distributions:** Beyond selfish mining, the risk of majority attacks by investigating shifts in miner share distributions is also studied [57]. The analysis examines the assumption that computational power is broadly distributed, i.e., no single entity should dominate the network, and proposes creating detailed

TABLE 4. Mining behavior anomaly detection.

Paper	Blockchain	Data source	Anomaly	Measure
[55]	BTC, LTC, ETH, BCH	On-chain (block)	selfish mining mining cartels	MSB-based p-values
[56]	BTC, LTC, ETH, BCH, MONA	On-chain (block)	selfish mining mining cartels	Type II binomial p-values
[57]	BTC, ETH	On-chain (block)	majority attack	N/A

profiles of each major miner or mining pool. By systematically tracking the evolution of these profiles over time, the approach flags anomalies indicative of rapid concentration of hash power, which elevates the threat of a 51% attack. Empirical findings on Bitcoin and Ethereum illustrate how abrupt spikes in a single miner's share function as early indicators of potential double-spending or extended block rewriting, offering a proactive means to detect and mitigate malicious consolidation of hashing resources.

These statistical techniques offer a harmonious blend of simplicity and sophistication in detecting anomalies across cryptoasset networks. Methods like Benford's Law and Mahalanobis-based scoring shine for their computational efficiency, ease of implementation, and broad generalizability across diverse datasets, while signature-based and tensor models, as well as forecasting frameworks, deliver deeper insights and capture complex temporal dynamics albeit at a higher computational cost. Although techniques based on basic distribution properties scale well and provide readily interpretable signals, more advanced approaches often require extensive parameter tuning and robust computational infrastructure, which can hinder real-time application and limit adaptability to rapidly evolving market conditions. Likewise, mining behavior anomaly detection methods effectively highlight irregularities in block discovery and pool dynamics but depend critically on accurate miner identification and are vulnerable to sophisticated adversarial strategies. Collectively, these approaches underscore a trade-off between simplicity and granularity, suggesting ample room for improvement through hybrid models, adaptive thresholding, and enhanced integration of external factors to bolster detection accuracy and scalability further.

B. NETWORK ANALYSIS

Network analysis approaches leverage the inherent graph structure of blockchain transaction networks to identify anomalies. These methods analyze structural properties, connectivity patterns, and topological features to detect suspicious behaviors that might indicate fraud, money laundering, or other malicious activities. For the brief theoretical aspect of graph construction and the relevant properties of the graph, refer to section II-E.

1) STRUCTURAL & COMMUNITY ANALYSIS

A notable body of literature concentrates on global network properties of blockchain transaction graphs, such as degree distributions, clustering, community structure, and core-periphery organization. These analyses frequently uncover unexpected hierarchies and densely connected communities of addresses, challenging the assumption that blockchains are fully decentralized. Moreover, detecting strongly clustered groups, short-lived ephemeral subgraphs, or community "islands" reveals that suspicious or anomalous activity may easily concentrate among a few addresses, underscoring how important these structural analyses are for security, compliance, and the overall health of blockchain ecosystems. A summary of the studies covered in this category is presented in table 5.

- **Cryptoasset Transaction Network Structure:** Several early works analyze the Bitcoin transaction network from a structural perspective, with [58] focusing on four years of transaction data and revealing a small-world topology. In such a topology, the average geodesic distance among addresses is quite short, implying a relatively high level of interconnectivity; however, high-degree hubs in these networks can act as de facto "centralized" nodes handling disproportionate transaction volumes, potentially undermining the ethos of full decentralization. Meanwhile, [59] and [60] explore broader Bitcoin data to show scale-free-like degree distributions in which a small minority of addresses dominate overall connectivity; thus, although path lengths remain short, control is concentrated among a handful of high-degree nodes. While scale-free behavior often arises in real-world systems, it raises concerns about single points of failure or suspicious concentration of network power; for example, a small clique of exchanges or mixers could become a structural choke point. Both studies further incorporate clustering and assortativity metrics, finding that the Bitcoin network is mildly disassortative: large hubs primarily interact with small nodes, forming star-like substructures centered on major exchanges or service addresses. These observations align with additional findings in [58], which notes that certain regions exhibit usage patterns heavily oriented around small-value gambling transactions, underscoring how socio-economic factors can foster specialized clusters of activity. These results highlight that although the Bitcoin network achieves short-path efficiency and maintains a degree of resilience, its reliance on a small number of hubs and the influence of regionally specific transaction behaviors can introduce potential vulnerabilities and undermine the cryptoasset's intended decentralization.
  Whereas the above focuses solely on Bitcoin, [61] compares a Bitcoin trader network and an adolescent friendship network using community detection and social network analysis techniques, revealing interesting parallels and distinctions. Both networks exhibit moderate clustering, meaning that nodes tend to form tightly-knit groups and some reciprocity. Reciprocity, in this context, refers to the tendency for relationships to be mutual, i.e., if one person or Bitcoin trader forms a connection with another, the other is likely to reciprocate the connection, creating a two-way relationship. It is also concluded that adolescents prefer a reciprocal relationship with the same gender and that drinkers tend to be more active in their social circle. Notably, this financial network displays assimilation rather than

homophily; users tend to trade more frequently within their own communities without a strong tendency to connect based on similar characteristics. Furthermore, unusually dense or exclusive subgroups in the Bitcoin network could serve as indicators of suspicious activity. Overall, these findings underscore the structural similarities and behavioral differences between social and financial networks, offering insights that are relevant for understanding dynamics in both domains.

While designed as a stablecoin bridging multiple exchanges, Tether has been studied from both community and global-structure standpoints. The study using a Social Network Analysis (SNA) of the Tether transaction graph [62] reveals that the Tether transaction graph lacks the small-world property, which typically characterizes robust and efficient networks; instead, large cryptoasset exchanges dominate the degree distribution, acting as central nodes with significant influence over transaction flow. Bitfinex emerges as a pivotal player due to its co-ownership and co-administration ties with Tether's issuer, exemplifying a "rich-get-richer" effect that suggests control by a few major entities, potentially enabling manipulative practices. The network's low assortativity, indicating that high-volume entities do not form stable links over time, points to transient periods of high trading activity rather than sustained market interactions. Additionally, the concept of "bubble networks," defined by short periods of intense trading centered on key nodes, mirrors financial bubbles and further highlights structural vulnerabilities.

- **Random Graph vs. Cryptoasset Transaction Graph:** Complementing these global analyses, [63] fits random-graph models, i.e., Chung–Lu [64] and Buckley–Osthus [65], to Bitcoin's structure by using mathematical frameworks that describe how edges form between nodes according to probabilistic rules and highlights the bowtie structure yet reveals that the data deviate from simple scale-free or random attachment models, exhibiting persistent anomalies such as over-centralized clusters and ephemeral spike subgraphs likely resulting from intentional participant behaviors, e.g., strategic transaction patterns or the use of mixing services, network evolution over time, and underlying economic forces, these deviations indicate that simple random models do not fully capture the network's structural features, with ephemeral subgraphs potentially representing abrupt transaction bursts or on-chain mixers that raise compliance concerns. In a similar direction, [66] employs random-walk embeddings for link prediction by modeling Ethereum transaction records as a Temporal Weighted Multidigraph (TWMDG), $G = (V, E)$, where V is the set of nodes (accounts) and E is the set of edges (transactions). Each edge e is represented as $e = (u, v, w, t)$, where u is the source node, v is the target node, w is the weight (transaction amount), and t is the timestamp. This model incorporates the temporal and weighted aspects of transactions, recognizing that the timing and size of transactions are important features. This temporal information models the network as evolving continuously over time with additions of links. Various random walk strategies then applied over the TWMDG, defining Temporal Successive Edges, $L_t(u) = \{e \mid Src(e) = u, T(e) \geq t\}$ as the set of edges leaving node u at or after time t and assigning selection probabilities $P_T(e)$ of the random walk selecting successive edge e at time t from node u would then be $P_T(e) = \frac{1}{|L_t(u)|}$ (unbiased) or with respect to timestamp or amount (biased). The results show that local features like degree alone are insufficient for uncovering hidden edges that form ephemeral or secretive transaction clusters, which may harbor potential money-laundering or consolidation strategies undetected when subgraph patterns are overlooked.
- **Hierarchical structures of Tokens:** Several studies on token networks and smart contracts explicitly demonstrate that nominally "decentralized" systems may exhibit pronounced core-periphery or hierarchical structures, thereby challenging the principle of network flatness. For example, [67] conducts community detection on the AAVE token transaction network on the Ethereum blockchain and reveals a dominant core comprising centralized exchanges, such as Coinbase and Binance, and key contract wallets that mediate most token flows. This concentration indicates that a small group of aggregator nodes can dominate transaction throughput, introducing single points of failure and potentially obscuring suspicious patterns like cyclical liquidity. Similarly, [68] confirms that removing a few top addresses, particularly major exchange accounts and

TABLE 5. Structural & Community analysis.

Paper	Blockchain	Data source	Anomaly
[58]	BTC	on-chain (transaction), off-chain (geo-locations)	hub-dominated
[59]	BTC	on-chain (transaction)	hub-dominated
[60]	BTC	on-chain (transaction)	hub-dominated
[61]	BTC	off-chain (social network, weighted trust of BTC)	hub-dominated
[62]	BTC (Tether)	on-chain (transaction)	hub-dominated
[63]	BTC	on-chain (transaction)	hub-dominated
[66]	ETH (EOA)	on-chain (transaction)	hub-dominated
[67]	ETH tokens (AAVE)	on-chain (transaction)	hub-dominated
[68]	ETH (EOA & token: AMP, BAT, DAI, UNI)	on-chain (transaction)	hub-dominated
[69]	ETH tokens (DAI, UNI, WBTC)	on-chain (transaction)	hub-dominated

pivotal smart contracts can fragment the connectivity of the entire Ethereum token network, underscoring a critical structural vulnerability. The study further employs the Jaccard Index $J(A, B) = \frac{|A \cap B|}{|A \cup B|}$, which quantifies the overlap in transaction patterns by comparing two sets of trading counterparts, with A and B representing, for example, the sets of counterparties that two different nodes interact with, and the Ordered Jaccard Index (OJI) $OJI(A, B) = \frac{|LCS(A,B)|}{|A \cup B|}$, where LCS denotes the longest common subsequence between two sets capturing sequential patterns in how accounts trade. Aside from raising security issues, such a structural vulnerability indicates that transactions are anything but uniformly distributed and might reflect a persistent risk if those key nodes are compromised or engage in manipulative behavior.

Finally, [69] extends insights into decentralized finance (DeFi) by analyzing transaction networks of three prominent Ethereum-based tokens—Dai (DAI), Uniswap (UNI), and Wrapped Bitcoin (WBTC)—using metrics such as diameter, modularity, and density. The analysis reveals centralized clusters bridging the network, where large exchanges and pivotal smart contracts act as intermediaries facilitating most transaction flows. These bridging nodes form cross-linked communities that both enhance liquidity by connecting isolated network segments and constrain transaction behaviors within specific clusters. This pattern suggests that, despite DeFi's decentralized branding, actual usage is dominated by a small set of heavily utilized addresses, potentially creating single points of trust or failure and exposing systemic vulnerabilities. Moreover, structural biases hint at hidden risks, such as coordinated market manipulation or irregular trading patterns, as modularity analysis uncovers clusters with high internal connectivity but limited external interactions, and centrality calculations highlight influential wallet addresses that critically shape market dynamics.

2) TEMPORAL & EVOLUTIONARY NETWORK METHODS

Whereas the previous subsection centers on static, cross-sectional analyses, the works below incorporate a time-based or evolutionary perspective, often investigating how blockchain transaction networks grow, shift, or correlate with external factors, e.g., exchange prices. Methodologically, they frequently deploy preferential attachment models, dynamic snapshots, or temporal embeddings, differentiating them from purely structural studies that do not track changes over time. A summary of the studies covered in this category is presented in table 6

- **Rich-Get-Richer:** Recently, various works have studied the preferential attachment, i.e., the "rich-get-richer" phenomenon in Bitcoin and Ethereum, each from a distinct lens. For example, [70] is among the earliest studies to show that Bitcoin's growth follows linear preferential attachment, i.e., new edges (transactions) arrive with probabilities proportional to existing node degrees or wealth. They showed that the probability of forming a new link connecting to the node v is
$$p(k_v) = \frac{k_v^{\alpha}}{\sum_w k_w^{\alpha}} \tag{23}$$
where k_v is the indegree of node v, and $\alpha \geq 0$. The probability that the new link connects to any node with degree k is
$$p(k) \propto n_k k^{\alpha} \tag{24}$$
Building on this, [71] relaxes the assumption of purely degree-based attachment by introducing node "fitness" η_i and the preferential attachment kernel as
$$A_{ij} = k_i^{\theta} k_j^{\theta} + \eta_i \eta_j \tag{25}$$
where for small θ the initial fitness differences are not significantly amplified, but for larger θ these differences can become prominent. The empirical results indicate that certain nodes persistently attract transactions because of higher fitness values, potentially overshadowing simpler linear-degree rules. Further extending these perspectives, [72] targets Ethereum tokens, showing that multiple ERC-20 networks display super-linear preferential attachment, indicating that a few nodes quickly become hubs. Complementarily, [73] synthesizes findings for both Bitcoin and Ethereum, confirming that hubs maintain their dominance even as overall market conditions and prices fluctuate. Deviations from expected preferential attachment can signal anomalies and potential fraud. For example, a sudden connection surge to a previously low-degree node or an unexpectedly high fitness score may raise suspicion.
- **Transaction network and Price Correlation:** Another stream of research addresses the temporal analysis of multiple cryptoassets or snapshots aligned with price variation. For instance, [16] identifies that the degree distribution of monthly transaction networks for Bitcoin, Ethereum, and Namecoin cannot be well-fitted by the famous power-law distribution, i.e., these networks exhibit heavy-tailed distributions rather than scale-free properties. This structural uniqueness is further emphasized by the observation that while both Bitcoin and Ethereum networks are heavy-tailed with disassortative mixing, where high-degree nodes connect to low-degree nodes, only Bitcoin exhibits small-world properties. These differences likely stem from Ethereum's diverse use cases, such as smart contracts, which create more complex transactional patterns than Bitcoin's simpler peer-to-peer transactions. Likewise, [74] uses weekly or daily transaction network snapshots of Bitcoin to show that during price drops, the network becomes more heterogeneous, i.e., dominant addresses continue trading while most users reduce activity,

amplifying market volatility. External shocks, such as the Mt.Gox bankruptcy, disrupted established patterns. Before Mt.Gox's collapse, the out-degree distribution where the probability that a node has k outgoing connections follows roughly $k^{-\alpha}$ was compatible with a power-law model in about 54% of snapshots. After Mt.Gox, this compatibility dropped to around 26%. This shift indicates a fundamental change in how users transact, reflecting a loss of confidence in centralized exchanges and a redistribution of activity across the network, thereby offering insights into the interplay between network structure and market trends.

Several studies also explicitly link dynamic network features to price forecasting or correlation. For instance, [75] applies Principal Component Analysis (PCA) to daily or weekly snapshots of Bitcoin's address-level graph, revealing that topological indicators such as concentration in node degrees can precede significant price shifts. Singular vectors derived from PCA show strong correlations with Bitcoin prices, suggesting that structural changes in the transaction network serve as reliable predictors. In a similar vein, [76] adopts correlation-tensor spectra for weekly XRP networks. A four-dimensional correlation tensor $C_{(i,j),(\alpha,\beta)}$ captures the relationships between different network features over time. To find the spectrum of the correlation tensor, a double singular value decomposition (DSVD) was applied to unfold the tensor C can be unfolded along a chosen mode (dimension):

$$C_{(i,j),(\alpha,\beta)} = U_{(i,j)}\Sigma_1 V^*_{(\alpha,\beta)} \tag{26}$$

$$V_{(\alpha,\beta)} = U_{(\alpha,\beta)}\Sigma_2 W^*_{(\alpha,\beta)} \tag{27}$$

Here, $U_{(\ldots)}$ and $V_{(\ldots)}$ are the left and right singular vectors, and Σ_i contains the singular values for the i-th mode. The first SVD in eq.26 is unfolded such that the (i, j) indices form the rows, and (α, β) become the columns while the second SVD in eq.27 unfold (α, β) in a manner analogous to the first SVD. In each SVD step, one obtains a list of singular values:

$$\Sigma_1 = \mathrm{diag}(\sigma_1, \sigma_2, \ldots),\ \ \Sigma_2 = \mathrm{diag}(\rho_1, \rho_2, \ldots) \tag{28}$$

where the largest overall singular values are obtained. The singular values represent the amount of variance captured by each corresponding singular vector. The largest singular values, found along the diagonal of each Σ_i, indicate the most significant patterns or modes of variation in that mode. These are used to identify which relationships between the network features impact XRP price movements most. The study discovers a distinctive relationship between the largest singular values and price peaks, offering early indicators for impending surges or drops. Extending these approaches further, [77] employs a partial-differential-equation (PDE) framework using time-varying chainlet patterns to model Bitcoin price fluctuations. Chainlets are small, directed pieces of the Bitcoin transaction network that capture common transaction patterns. The idea is that each group (or cluster) of similar chainlets has a certain influence on the Bitcoin price, which we denote by $u(x, t)$ where x represents an abstract position that orders the chainlet clusters, i.e., clusters with similar transaction patterns are placed close together. The PDE framework uses these chainlets to model the continuous evolution of Bitcoin price movements:

TABLE 6. Temporal & Evolutionary network methods.

Paper	Blockchain	Data source	Anomaly
[70]	BTC	on-chain (transaction)	hub-dominated (rich-get-richer)
[71]	BTC	on-chain (transaction)	preferential attachment
[72]	ETH tokens (USDT, Chain-link,Binance coin)	on-chain (transaction)	preferential attachment
[73]	BTC and ETH	on-chain (transaction)	preferential attachment
[16]	BTC, ETH and Namecoin	on-chain (transaction)	hub-dominated
[74]	BTC	on-chain (transaction)	network structure correlated with price (Mt.Gox)
[75]	BTC	on-chain (transaction)	network structure correlated with price
[76]	XRP	on-chain (transaction)	network structure correlated with price
[77]	BTC	on-chain (transaction)	modeling price movement using network structural properties
[78]	BTC	off-chain (payment channel)	lightning network
[79]	ETH	on-chain (transaction)	ponzi and phish

$$\frac{\partial u(x,t)}{\partial t} = \frac{\partial}{\partial x}(d(x)\frac{\partial u(x,t)}{\partial x}) + r(t)u(x,t)h(x) \tag{29}$$

The term $\frac{\partial}{\partial x}(d(x)\frac{\partial u(x,t)}{\partial x})$ models the diffusion of influence across clusters, with $d(x)$ describing how interactions vary spatially and the term $r(t)u(x, t)h(x)$ captures the local growth or decay of this influence. The study concludes that expansions or contractions within transaction subgraphs act as short-horizon signals for bull or bear dynamics.

- **Temporal Change of Transaction Network:** One specialized approach is found in [78], which measures the Lightning Network's growth to test if it follows a Barabási–Albert (BA) scale-free pattern. The BA model generates networks where a few nodes are highly connected hubs due to preferential attachment; new nodes connect to existing nodes with high degree, following a power-law degree distribution. Their analysis of newly opened channels reveals that the network deviates from the pure BA model. Specifically, new nodes tend to connect to existing nodes with greater Closeness Centrality rather than simply connecting to high-degree nodes as predicted by the BA model. This suggests that nodes are strategically choosing connections to enhance routing performance within the Lightning Network rather than simply maximizing their number of connections, implying that the BA model may not be optimal for simulating or designing routing protocols for the Lightning Network.

Rather than analyzing the entire network at once, [79] focuses on locally dynamic structures by building ego networks for labeled Ethereum accounts (e.g., ICO, Mining, Gambling, Ponzi). Ego networks are subgraphs centered on a single node (the "ego") that includes its immediate neighbors (the "alters") and all the connections among those neighbors. This approach provides a localized view of an account's direct transaction environment and captures dynamic, micro-level interactions that can be obscured in a global analysis. The study finds that illegal accounts (Ponzi and Phish) have much shorter lifecycles (less than 20 days) compared to normal accounts. It also reveals that ICO accounts exhibit high local clustering ($\approx$ 0.18), suggesting that ICO investors frequently transact with one another, while gambling accounts have very low clustering ($\approx$ 0.024), reflecting their sporadic interaction patterns. Furthermore, the ratio of in- to out-transactions varies by account type, and mining, exchange, and Ponzi accounts show a higher proportion of out-transactions, which reflects their distinct operational roles.

3) GRAPH-BASED DETECTION & DE-ANONYMIZATION

Whereas the previous subsections emphasize either static structure or temporal evolution, the studies below deploy graph-based methods to uncover malicious usage, suspicious flows, or anonymity breakdown in blockchain transaction networks. These methods often involve refined address-clustering heuristics, subgraph-based anomaly detection, or specialized modeling techniques, enabling the identification of fraudulent behavior. A summary of the studies covered in this category is presented in table 7

- **Address Clustering & De-Anonymization:** A key theme is the use of enhanced address clustering to reveal hidden links and partially de-identify actors. For example, [80] focuses on Zcash, an altcoin of Bitcoin aiming to protect blockchain anonymity, extending the traditional multi-input heuristics, which assume that all input addresses in a transaction belong to the same user by combining with tracking change (shadow) addresses, addresses automatically generated to return leftover funds from a transaction to the sender. This results in an increase in the clustering rate by 9% as the change addresses often belong to the same user as the input addresses, providing a way to link transactions and cluster addresses more effectively. Despite zero-knowledge proofs, repetitive spending patterns like round-trip transactions can partially deanonymize activity, with 87.5% of addresses and 25.7% of transactions linked to mining rewards and shielded pools used mainly by founders, miners and mining pools rather than typical privacy-focused users. Similarly, [81] explores the Bitcoin network into entities such as exchanges, gambling sites, and miners using features like multi-input patterns and transaction rates, which further refine classification by analyzing behavioral trends over time. These features are used in a classification method that applies clustering algorithms and statistical analysis to group addresses into entities with consistent behavior patterns. This allows outliers, i.e., addresses exhibiting unexpected behaviors, to be flagged as suspicious.
- **Transaction Flow & Anomaly Analysis:** Several studies address manipulative or fraudulent behaviors in cryptoasset markets. Reference [19] analyzes leaked Mt. Gox data to reveal potential price manipulation linked to abnormal trading activity by constructing user-level transaction graphs. Accounts involved in "extremely high" and "extremely low" transactions, those significantly deviating from the average market price on a given day, are identified. These abnormal accounts (ABA), which are classified into extremely high accounts (EHA) and extremely low accounts (ELA), represent 12.5% of the accounts and approximately 2.8% of transactions with ABA accounts. These abnormal accounts were correlated with sudden price changes via SVD, where transaction data are first divided into daily snapshots and then represented these snapshots as matrices. Then, SVD was applied to extract "base networks," i.e., dominant patterns of transactions within the network. The results show that the abnormal accounts transactions strongly related to the Bitcoin price especially the volume and direction of transactions involving EHAs and ELAs, significantly correlated with fluctuations in the Bitcoin price on Mt. Gox. Similarly, [82] proposes a Petri-net–based framework to model concurrency and dynamic transaction flows in Bitcoin. The model extracts nineteen transaction features. For example, the in/out ratio measures the balance between incoming and outgoing transactions for an address, where a high in/out ratio may indicate an accumulation phase, while a low ratio suggests funds are being drained. The identification of short cycles, where funds rapidly move through a series of addresses and return to the origin, can be indicative of layering techniques used to obscure the source of funds. By combining these features, the framework aims to provide a more comprehensive approach to blockchain forensics.

  Concerning computational performance, [83] focuses on GPU-accelerated methods for subgraph-based anomaly detection to address the computational challenges of

TABLE 7. Graph-based detection & De-anonymization.

Paper	Blockchain	Data source	Anomaly
[80]	BTC (Zcash)	on-chain (transaction)	deanonymization
[81]	BTC (Zcash)	on-chain (transaction)	user behaviour
[19]	BTC	on-chain (transaction)	fraudulent (Mt.Gox)
[82]	BTC	on-chain (transaction)	fraudulent (Mt.Gox)
[83]	BTC	on-chain (transaction)	fraudulent detection using GPU
[84]	ETH	on-chain (transaction)	money laundering
[85]	ETH	on-chain (transaction)	phishing activities
[86]	EOSIO	on-chain (transaction)	fraudulent and bot-like accounts
[87]	EOSIO	on-chain (transaction)	scam behavior

analyzing large datasets. By constructing localized subgraphs around each target transaction and analyzing them with outlier-detection algorithms, the method is scalable by leveraging the parallel processing capabilities of GPUs, making it feasible to analyze large datasets while maintaining effectiveness in identifying anomalous transactions. For Ethereum, [84] investigates transactions from an alleged Upbit exchange hack to study on-chain laundering patterns. A money laundering network on Ethereum was constructed by crawling the transaction records of the Upbit Hack and then conducting an analysis of the money laundering network properties by comparing the money laundering network with the normal network on Ethereum. Despite Ethereum's fast transaction capabilities, the results show that money laundering accounts on Ethereum are fast-in and fast-out accounts, meaning that dirty money is transferred in and out quickly by money laundering accounts. Also, compared with traditional money laundering accounts that usually transfer high-volume money, prudent money laundering accounts on Ethereum tend to transfer very small-volume money to evade the attention of regulatory authorities. The actors take advantage of decentralized exchanges for rapid layering to obscure the origin of funds and evade detection. They also found that, like traditional money laundering accounts, money laundering accounts on Ethereum are zero out middle accounts, meaning that they potentially transfer almost all incoming money out to benefit in a big way.

- **Subgraph Patterns:** Meanwhile, [85] introduces Transaction SubGraph Networks (TSGN) for phishing detection. The study used embed local subgraphs around potentially malicious addresses and observe that ephemeral, cyclical in–out flows are reliable indicators of scam behavior. In phishing attacks, funds typically flow into the scammer's address and are quickly moved out through a series of transactions to obscure their origin. The presence of cyclical flows, where funds return to addresses controlled by the attacker, further indicates coordinated fraudulent activity. The TSGN approach effectively identifies phishing scams on the Ethereum network by focusing on these subgraph patterns.
  Recent investigations into the EOSIO blockchain reveal that even systems with high transaction throughput remain vulnerable to systematic manipulation. Transaction-graph analytics are applied to uncover that a significant portion of accounts exhibit bot-like behavior. For instance, [86] analyzes features such as the time intervals between transactions and bursty co-activity where multiple accounts perform actions in close temporal proximity to identify accounts with regular, predictable patterns indicative of automation. Their analysis reveals that over 30.75% of the accounts (381,008 in total) exhibit bot-like behavior, participating in more than 192 million transactions and transferring around 640 million EOS tokens in repetitive and exploitative ways for malicious purposes like bonus hunting and clicking fraud. Similarly, [87] leverages local subgraph embeddings around potentially malicious addresses and observes that short-lived, recurrent transaction cycles are reliable indicators of scam behavior. By combining these various features, accounts that are systematically abusing the high-throughput capabilities of EOSIO are identified.

Although these graph-based methodologies have yielded rich insights into blockchain networks, several important considerations remain. Noted that Table 5, 6 and 7 do not contain a "Measure" column since these network-focused methods predominantly emphasize topological or structural features of the transaction graph rather than, for instance, specific predictive or classification metrics. On the plus side, structural or community analyses effectively illuminate how small sets of aggregator nodes or hub addresses exert large-scale influence, and they are readily generalized to different cryptoassets or token systems by simply redefining node or edge types. Temporal and evolutionary approaches—such as dynamic snapshots, preferential-attachment models, or subgraph anomaly detection—add further realism and can capture short-lived or bursty behaviors often missed by purely static analyses. However, all of these techniques face scalability challenges as blockchains grow in both transaction volume and participant diversity, and complex subgraph or multi-dimensional embeddings can quickly become computationally expensive for large datasets. Furthermore, clustering heuristics and models like Chung–Lu or Barabási–Albert can fail to capture special nodes (e.g., centralized exchanges, mixers) or ephemeral patterns arising from purposeful on-chain manipulations, limiting their predictive power. Real-world heterogeneities such as multi-signature addresses, advanced DeFi operations, or bridging solutions spanning multiple blockchains complicate straightforward generalization. Moreover, while local subgraph extraction helps isolate suspicious flows, it risks overlooking broader interactions that cross these local boundaries. Hence, future improvements might focus on more scalable high-performance computing (e.g., GPU-based pipelines) plus adaptive heuristics that incorporate domain-specific behaviors (mixers, privacy protocols, exchange deposit wallets, etc.) while balancing interpretability with the complexity required to handle blockchains' rapidly shifting topologies.

C. MACHINE LEARNING

Machine learning approaches have become increasingly dominant in cryptoasset anomaly detection due to their ability to learn complex patterns from large-scale transaction data. These methods can be categorized based on their learning paradigm and architectural design.

1) SUPERVISED LEARNING

Supervised learning methods rely on labeled datasets to train models that can classify transactions or addresses as legitimate or anomalous. A number of works have applied classical supervised ML for cryptoasset fraud detection, focusing on constructing domain-specific features and training algorithms such as Random Forest, SVM, LightGBM, or XGBoost. Feature engineering plays a crucial role in model performance, with successful approaches incorporating features from multiple dimensions ranging from raw transaction records to abstract topological or temporal metrics consistently boosting detection accuracy. The most effective supervised methods incorporate diverse feature types, typically encompassing (i) **Transaction features** such as amount, fee, timestamp, and confirmation time, (ii) **Temporal features** such as transaction frequency, timing patterns, and burst behavior, (iii) **Graph features** such as in/out degree, clustering coefficient, centralities, and (iv) **Behavioral features** such as address reuse, transaction size distribution.

- **Fraud & Suspicious Activity Detection:** Several studies adopt classic ML approaches with carefully engineered features. In [88], a combination of Random Forest and XGBoost is employed for fraud detection in Bitcoin. High accuracy is achieved by synthesizing transaction and graph-based metrics, e.g. transaction amounts, node degrees, and edge timestamps, allowing the ensemble to capture both local (transaction-level) and global (address-level) patterns. Similarly, [89] proposes a stacking ensemble using decision trees, naive Bayes, k-nearest neighbors, and random forest for Bitcoin fraud. Adaptive Synthetic Sampling (ADASYN) is utilized to address class imbalance, complemented by SHAP for interpretability. ADASYN is an oversampling technique that generates synthetic samples for the minority class (e.g., fraudulent transactions) by focusing on harder-to-learn examples. Unlike simpler oversampling methods like SMOTE, ADASYN assigns higher weights to minority class samples that are harder to classify, thereby improving the model's ability to distinguish between legitimate and fraudulent transactions. An F1-score above 95% reflects the synergy between robust feature engineering including user-specific transaction frequency and connectivity and ensemble-based model fusion.
Recent work in Ethereum phishing and suspicious address detection leverages a variety of machine learning techniques and feature engineering approaches. For instance, [90] uses XGBoost and RF with a blend of structural and temporal attributes in the Ethereum transaction network. By quantifying transaction frequency, inter-event timing, local node degrees, and address reuse, their pipeline achieves 98% F1-scores for phishing detection. Meanwhile, [91] focuses on node2vec embeddings combined with Adaptive Boosting (AdaBoost) to detect money laundering in Bitcoin, concluding that temporal behaviors and graph-based embeddings rank among the most important features. Likewise, [92] presents "GuiltyWalker," a method that measures each address's distance from known illicit nodes via random walks; these distance-based features, when fed into RF yield notable accuracy gains for malicious address identification. Across these studies, consistent improvements arise from layering graph connectivity features, e.g. in/out degrees, clustering coefficients, over the more common transaction or temporal signals.
- **Ponzi Scheme & HYIP Identification:** Other works target specific subproblems, such as Ponzi scheme detection. Early examples include [93] and [94], which rely on SVM, decision trees, and XGBoost to detect Ponzi contracts on Ethereum. Results demonstrate that combining smart contract code-level signals, e.g., extracted opcodes and function usage, with transaction-based metrics, i.e., frequency and daily volume, measurably improve classifier precision and recall. In line with this, [95] applies standard text classification using SVM and Naive Bayes on Solidity code tokens for Ponzi detection. This approach treats the smart contract code as text and utilizes natural language processing methods to identify patterns indicative of Ponzi schemes. The near-perfect accuracy reported with 99% overall accuracy underscores that raw contract text, featuring code usage patterns and address references, can effectively signal suspicious activity. This indicates that even without a deep analysis of the opcodes or transaction history, the textual content of the contract itself contains discriminative features that can be used to detect Ponzi schemes. Building upon this line of work, [96] proposes heterogeneous feature augmentation (HFAug), a feature augmentation scheme that integrates heterogeneous transaction records, e.g., transaction amounts, time lags between consecutive transactions and frequency of transactions, and meta-path-based structural features. When evaluated using Logistic Regression (LR), SVM,

and RF, results confirm that capturing both temporal and graph structures significantly strengthens classification performance for Ponzi detection.

Focusing on Ponzi or High-Yield Investment Program (HYIP) detection, [97] uses RF, NN, and k-NN to detect Ponzi schemes on Ethereum. Over 20,000 Ethereum transactions were analyzed and preprocessed to train the models. Their main result shows that a large, over 70 sets of raw features can be pruned down to about 10 core features without compromising accuracy. These core features likely include transaction-level and address-level metrics, such as transaction amounts, frequency, timing intervals, and patterns indicative of Ponzi schemes. Similarly, [98] tackles the identification of HYIP operators' Bitcoin addresses via a custom scraping-based approach. They highlight the effect of transaction features like frequency of transactions per day, deposit–withdrawal patterns, and transaction size distributions on classification performance, concluding that gleaning large labeled sets is critical to robust supervised detection.

- **Address Role Classification & Scalable Pipelines:** Another group deals with GPU-accelerated or large-scale supervised pipelines. References [99] and [100] adopt SVM, RF, and Logistic Regression on tens of millions of Bitcoin/Ethereum transactions, showcasing that parallelization (e.g., GPU computing) is essential for near-real-time detection. Their data includes advanced features like node centralities, transaction bursts, and timing intervals, and the results indicate that even incremental improvements in feature engineering can manifest as large gains in detection speed and precision on these large networks. Similarly, [101] examines suspicious-user detection in Bitcoin trust networks with RF, deriving especially strong signals from node centralities and trust-based features, where users rate each other on a scale to indicate their level of trust, which capture how user reputation, quantified through the trust scores assigned to the user by their peers, and transaction patterns connect.

  Recent works highlight role classification rather than direct anomaly detection. Reference [102] trains RF and XGBoost to classify Ethereum addresses as exchanges, wallets, or other key agents. They show that addresses exhibit distinctive transaction frequencies and code usage patterns, making assigning roles with high confidence feasible. Extending this, [103] introduced a pipeline called GTN2vec to embed Ethereum addresses with features like gas price and timestamps in random walks, enabling robust money laundering detection via RF classifiers. Similarly, Bitcoin-focused studies have developed specialized approaches for address classification. Reference [104] proposed moment-based features such as variance and skewness of transaction amounts and used LightGBM to achieve high F1 scores for abnormal address detection. This was further expanded by [105] to include multi-digraph embeddings that incorporate transaction time windows, highlighting the importance of temporal features and burst behaviors in enriching graph-based signals. Expanding on these efforts, [106] introduces XGBCLUS, a framework designed for anomaly detection that combines XGBoost with under-sampling techniques to address class imbalance to detect anomalies such as fraudulent or malicious activities within Bitcoin networks. By integrating explainable AI techniques like SHAP, the results show how features such as transaction volumes play a paramount role in classifying anomalous transactions.
- **Supervised Deep Learning Applications:** Lastly, [107] demonstrates a supervised deep-learning approach that uses an LSTM/Bi-LSTM/CNN ensemble for Ethereum phishing classification. Although these are indeed deep neural architectures, the pipeline is fully supervised, relying on a labeled dataset of malicious and benign addresses. Contrary to some graph-based methods, the authors do not incorporate domain-specific features (e.g., transaction frequency, node degrees, or gas usage). Instead, they embedded the raw addresses and fed them into the ensemble model, achieving near 99% detection accuracy. This outcome underscores the strength of combining address-level embeddings with advanced neural networks for phishing detection in Ethereum.

Table 8 provides an overview of representative supervised techniques for cryptoasset anomaly detection, highlighting their performance metrics, data sources, and target anomalies. In general, Random Forest (RF) appears frequently and often outperforms other classic ML methods, e.g., decision trees, SVM, or logistic regression, likely due to its robustness against noisy features and its ability to capture both nonlinear and interaction effects among transaction, temporal, and graph inputs. However, a major drawback of most supervised approaches is their susceptibility to class imbalance, as many real-world datasets exhibit far fewer fraudulent or malicious samples than legitimate ones. Although techniques like SMOTE or ADASYN partially address this imbalance, oversampling can introduce synthetic noise, while undersampling risks discarding informative samples. Moreover, many of these studies rely heavily on public, on-chain datasets, which may omit off-chain data such as user reputations or external intelligence. Methods that exploit private or proprietary data like trust scores, code annotations, or exchange user logs may improve accuracy but are less generalizable if these proprietary sources are not publicly available. Finally, while ensemble and deep-learning pipelines can be scaled to large transaction networks (some employing GPU acceleration for tens of millions of records), their performance may still be constrained by the quality and consistency of labels, underlining the continuing importance of robust data collection and labeling strategies for new research directions.

TABLE 8. Supervised learning methods.

Paper	Techniques	Blockchain	Data source	Anomaly	Measure
[88]	XGBoost, RF	BTC	On-chain (transaction)	fraudulent	AUC:0.92
[89]	XGBoost, decision trees, naive Bayes, k-NN,SHAP	BTC	On-chain (transaction)	fraudulent	AUC:0.99, F1:0.97, precision:0.96, recall:0.98
[90]	LightGBM, RF, MLP, SVM, logistic regression, XG-,Ada-, Cat Boost	ETH	On-chain (transaction)	phishing scams	F1:0.98
[91]	Adaboost	BTC	On-chain (transaction)	money laundering	accuracy:0.92, F1:0.93
[92]	RF	BTC	On-chain (transaction)	money laundering	F1:0.85, recall:0.77, precision:0.97
[93]	XGBoost	ETH	On-chain (transaction)	Ponzi schemes	F1:0.86, recall:0.81, precision:0.94
[94]	XGBoost, SVM, RF, decision trees	ETH	On-chain (smart contract code)	Ponzi schemes	F1:0.79, recall:0.69, precision:0.95
[95]	SVM, naive Bayes, decision trees	ETH	On-chain (smart contract code)	Ponzi schemes	accuracy:0.99, AUC:0.99
[96]	SVM, RF, GNN, logistic regression	ETH	On-chain (smart contract transaction)	Ponzi schemes	F1:0.86
[97]	RF, NN, k-NN, SVM, logistic regression	ETH	On-chain (transaction)	Ponzi schemes	precision:0.94, F1:0.94, recall:0.94
[98]	RF, XGBoost, NN, SVM, k-NN	BTC	On-chain (transaction)	HYIP, Ponzi schemes	accuracy:0.93
[99]	RF, SVM, logistic regression	BTC , ETH	On-chain (transaction)	fraudulent	accuracy:0.97, recall:0.98
[100]	RF, SVM, XGBoost, MLP	BTC	On-chain (transaction)	fraudulent	precision:0.98, recall:0.80, F1:0.86
[101]	RF, SVM, MLP, DT, naive Bayes	BTC	On-chain (transaction), Off-chain (trust scores)	user behaviour	precision:0.93, recall:0.94, F1:0.93
[102]	LightGBM, SVM, MLP, RF, logistic regression	ETH	On-chain (transaction)	user behaviour	accuracy:0.89, recall:0.86, F1:0.86
[103]	XGBoost, SVM, logistic regression, RF, naive Bayes	ETH	On-chain (transaction)	money laundering	accuracy:0.95, recall:0.96, F1:0.96
[104]	XGBoost, MLP, logistic regression, RF, naive Bayes, SVM, LightGBM	BTC	On-chain (transaction)	user behaviour	F1:0.87
[105]	SVM	ETH	On-chain (transaction)	phishing attacks	precision:0.85, recall:0.83, F1:0.85
[106]	XGBoost, DT, RF, AdaBoost	BTC	On-chain (transaction)	fraudulent	accuracy:0.97, AUC:0.92
[107]	LSTM, Bi-LSTM, CNN-LSTM	ETH	On-chain (transaction)	phishing attacks	accuracy:0.99

2) UNSUPERVISED AND SEMI-SUPERVISED LEARNING

Unsupervised learning methods address the challenge of limited labeled data in cryptoasset ecosystems by identifying intrinsic patterns without requiring extensive ground-truth annotations. These approaches aim to capture the underlying structure in transaction or address networks, allowing researchers to flag suspicious or outlier behaviors.

- **Clustering-Based Anomaly Detection:** A number of studies focus on clustering-based techniques to separate normal versus anomalous user activity. For instance, [108] adopts trimmed k-means on Bitcoin data to isolate potential fraud clusters by removing outliers that might distort the centroids. Their experiments demonstrate that removing a small percentage of extreme points before clustering significantly improves overall fraud detection rates. Similarly, [109] and [110] combine k-means, Mahalanobis distance, and unsupervised Support Vector Machines (SVMs) to detect anomalies in both user-centric and transaction-centric graphs. For a user-centric graph, each node represents an individual user, aggregating one or more Bitcoin addresses, and edges between nodes capture transactions between users. In contrast, a transaction-centric graph treats each transaction as a node, and edges typically represent the Bitcoin flow. By extracting features such as in-degree, out-degree, average transaction size, and time-interval statistics, their pipelines reveal that suspicious transactions generally deviate markedly from the typical distribution of user behavior. Meanwhile, [111] proposes a two-stage approach where One-Class SVM first flags outliers among Bitcoin transactions, then k-means groups similar outliers by type of attack (e.g. double-spending, malicious campaigns). This dual-step pipeline improves interpretability, as each cluster of anomalies is mapped to likely fraud scenarios.
- **Collective & Address Aggregation Approaches:** Other works focus on either addressing large-scale or complex transaction graphs. Reference [112] studies malicious address identification in Bitcoin by combining temporal burst features, e.g., abrupt increases in transaction volume or degree, and graph-based metrics, e.g., clustering coefficient, in/out-degree. The study highlights that aggregating addresses controlled by the same user is crucial for achieving more accurate anomaly scoring, i.e., disregarding the concept of "change addresses" can dilute signals indicative of malicious behavior. In a typical Bitcoin transaction, a user must spend the entire input, even if they intend to send only a part of that amount to another party. The remaining balance is then sent back to the sender's wallet via a new, often unrelated-looking address called a change address. Reference [113] extends such ideas using a collective anomaly detection paradigm in Bitcoin, whereby clusters of wallets owned by the same user are analyzed as a whole rather than individually. Experimental results show that considering the joint behavior across multiple addresses can increase recall in identifying malicious or hacked accounts since fraudsters often split illicit funds among numerous addresses.

- **Semi-Supervised Learning with Graph Embeddings:** Beyond direct clustering, some semi-supervised approaches leverage graph embeddings or node representation learning to detect scams. For example, [114] implements a network embedding pipeline for Ethereum phishing detection by incorporating transaction metadata (e.g., amount, timestamp). After embedding addresses into a low-dimensional space, they apply one-class SVM to separate normal from phishing nodes. Results indicate that preserving both temporal and weighted edge information during embedding (transaction sums, frequency) can markedly enhance the recall for phishing address detection, highlighting that more nuanced embeddings capture subtle fraudulent signatures more effectively than simpler topological embeddings alone.

Unsupervised and semi-supervised methods (Table 9) address the lack of large labeled datasets by detecting intrinsic structure or outliers within blockchain transaction networks, a clear advantage over fully supervised approaches that require extensive annotation. Because many suspicious behaviors are subtle or evolve quickly, these clustering and outlier-based techniques, e.g., k-means and One-Class SVM, excel at capturing new or emerging fraud patterns that strictly supervised pipelines might miss. Moreover, grouping suspicious actors without prior labels provides a practical first step in highlighting high-risk users or transactions for subsequent investigation. By detecting the intrinsic structure or outliers within the network, these methods mitigate the imbalance problem inherent in scarce labeled data, a limitation that fully supervised approaches must contend with. However, these methods often suffer from limited interpretability, e.g., why a cluster is flagged as suspicious can be unclear, and false positives may be high without further refinements, such as combining domain-specific features or post-hoc classification to filter alerts. Compared to the fully supervised approaches discussed in the previous section, which tend to achieve higher precision given abundant labeled data, unsupervised pipelines must carefully tune hyperparameters, e.g., number of clusters, outlier thresholds, and incorporate domain knowledge, e.g., change address heuristics, to reduce noise. Consequently, while these approaches are immensely flexible and scalable for preliminary screening or for newly emerging fraud vectors, practitioners may ultimately need to fuse them with supervised classifiers, where labels are available, to maximize detection performance and interpretability.

3) DEEP LEARNING & GRAPH NEURAL NETWORKS

Deep learning architectures have demonstrated exceptional performance in cryptoasset anomaly detection by automatically learning hierarchical representations from raw transaction data. Neural network approaches such as multilayer perceptrons (MLPs), convolutional neural networks (CNNs), and recurrent neural networks (RNNs) have been widely applied to tasks ranging from suspicious address classification to contract-level fraud detection.

TABLE 9. Unsupervised and semi-supervised learning methods.

Paper	Techniques	Blockchain	Data source	Anomaly	Measure
[108]	k-means clustering (Classical and trimmed)	BTC	On-chain (transaction)	fraudulent	Detecting 5 out of 30 known fraud cases. No standard classification metrics applied.
[109]	k-means clustering, SVMs	BTC	On-chain (transaction)	user behaviour	N/A
[110]	k-means clustering, SVMs	BTC	On-chain (transaction)	user behaviour	N/A
[111]	k-means clustering, SVM	BTC	On-chain (transaction)	DDoS, double spending, 51% attacks	accuracy:0.9
[112]	k-means clustering	BTC	On-chain (transaction)	fraudulent	N/A
[113]	trimmed k-means clustering	BTC	On-chain (transaction)	fraudulent	detected 14 users involved in 9 distinct cases of frauds
[114]	SVM, logistic regression, naive Bayes	ETH	On-chain (transaction)	phishing attacks	precision:0.92, recall:0.89, F1:0.9

- **Temporal GNNs & Dynamic Analysis:** Multiple studies leverage time-evolving behavior in transaction records. In [115], a temporal GCN is integrated with an LSTM backbone to detect illicit Bitcoin transactions by capturing dynamic changes in the Elliptic dataset. Exploiting the chronological ordering of transaction blocks enhances classification accuracy, as each block includes a timestamp indicating when it was mined, forming an ordered sequence $B_k \rightarrow B_{k+1} \rightarrow B_{k+2} \ldots$ that reflects the timeline of transaction appearance on the ledger. For example, if block k is followed by block $k+1$, the transactions in block $k+1$ must happen after those in block k. By treating each block as a temporal slice, the model identifies evolving patterns, e.g., unusual transaction values or addresses, rather than assuming all transactions occur simultaneously. Relatedly, [116] constructs forward and reverse Ethereum transaction graphs and applies a bi-graph attention-based network (LB-GLAT) to address the limitations posed by the acyclic nature of transaction graphs, which can obscure contextual relationships. The forward graph captures the natural flow of funds from senders to receivers, while the reverse graph, constructed by inverting edge directions, reveals the origin of funds. Learning from both directions improves the detection of money laundering. Reference [117] formalizes the detection of malicious Ethereum activity using multi-layer temporal snapshots across multiple tokens. Their approach integrates these snapshots within a temporal framework by segmenting the transaction data into distinct time windows based on the timestamp on each transaction. Snapshots from different tokens that fall within the same time window are merged into unified graphs, to which a graph convolution encoder is applied to extract spatial and temporal features. This enables the model to effectively capture cross-token trading patterns and detect evolving behaviors, such as sudden shifts in transaction volumes or unusual flows that may indicate malicious activities. The model significantly improves precision and recall by integrating these temporal snapshots with a GNN-based encoder. In a related approach, [118] uses a time-decayed mechanism to build dynamic transaction subgraphs for Bitcoin forecasting (DLForecast). The results show that weighting recent transactions more heavily substantially boosts accuracy in predicting future edges (transactions) and highlights potential anomalies earlier.

- **Transformer-Based Approaches:** Transformer models, which leverage self-attention mechanisms to capture relationships between elements in sequences, have been widely adopted for anomaly detection due to their capacity to process long sequences and extract complex patterns from unstructured data. Reference [119] propose BERT4ETH, a pre-trained Transformer-based model that treats sequences of Ethereum addresses and transactions as "tokens" within a language-modeling framework. In natural language processing, tokens typically represent words or subwords that serve as the fundamental units for learning representations. In this context, a subset of addresses in a transaction sequence is randomly replaced with a special [MASK] token, and the model is trained to predict the masked addresses using the surrounding unmasked tokens. This masked modeling strategy encourages learning robust contextual relationships among addresses, yielding notable improvements in tasks such as phishing classification and de-anonymization. Similarly, [120] integrates a Variational Autoencoder (VAE) with a Transformer architecture to detect anomalies in decentralized finance (DeFi) protocols. The VAE compresses data into a low-dimensional latent space while preserving local features, effectively capturing short-term behavioral patterns within limited time windows. In contrast, the Transformer component models long-range dependencies, enabling the detection of relationships between temporally distant events. These long-range dependencies enhance the model's ability to detect patterns where past behaviors influence future activity. The resulting framework, Anomaly VAE-Transformer, demonstrates strong performance in identifying malicious structural shifts, such as those associated with flash-loan attacks, and outperforms conventional CNN- and LSTM-based methods on large-scale DeFi datasets.
- **Heterogeneous, Multi-View & Subgraph-Focused GNNs:** Some approaches emphasize the use of multi-type edges or multi-view channels in transaction networks. Multi-type edges reflect that not all relationships in a transaction graph are homogeneous; edges may represent distinct types of interactions, for instance, a basic fund transfer versus a contract code invocation, or correspond to different analytical perspectives. This idea is often operationalized through multi-view channels, where each channel represents a subgraph that captures a specific facet of the overall network. Reference [121] uses a heterogeneous graph neural network based on a relational graph convolutional network (RGCN) to account for diverse transaction types on Ethereum, such as contract calls and standard transfers. Explicitly modeling each edge type by assigning distinct parameters to different transaction categories proves crucial for effective phishing detection, particularly in scenarios with label imbalance. Meanwhile, [122] integrates Bayesian uncertainty modeling with a multi-channel graph attention network to secure Ethereum-based Internet of Things (IoT) transactions. Incorporating Bayesian uncertainty enables a more robust handling of noise and class imbalance by allowing prediction adjustments based on estimated uncertainty. The multi-channel aggregator processes different transaction subgraphs independently, improving robustness and classification performance when identifying anomalous IoT device addresses.

  Methods proposed in [123] and [124] emphasize the importance of extracting localized subgraphs around target addresses for improved classification. Zhou et al. [123] introduce Ethident, a hierarchical GNN (HGATE) framework that samples micro interaction subgraphs from Ethereum and conducts classification at the subgraph level. To address label scarcity, a contrastive self-supervision module is incorporated, resulting in a 1–5% relative improvement in accuracy compared to baseline GNN models. In a complementary line of work, Nicholls et al. [124] propose FraudLens, which restructures the Bitcoin transaction graph through affinity- or feature-based edge construction prior to GNN training. Refinement of the graph structure through the removal of extraneous edges leads to substantial gains in classification performance when identifying illicit transaction nodes.

  Several studies adopt a star-shaped subgraph centered around each suspicious address. Reference [125] focus on phishing detection by constructing star subgraphs enriched with multi-scale features, including inbound and outbound transaction volumes, node lifetime, and other relevant attributes. The resulting GNN-based classification achieves nearly 99% recall on phishing-labeled addresses, effectively capturing localized transactional patterns characteristic of phishing activity.

Likewise, [126] also utilizes star subgraphs but emphasizes the aggregation of both node and edge features through a two-layer attention mechanism. By incorporating manually engineered features—such as minimum and maximum transaction values—into the node embeddings, the approach significantly enhances detection performance, reaching up to 99.3% recall. This represents a substantial improvement over embedding-only baselines such as DeepWalk. In the same vein, [127] proposes MP-GCN for phishing node identification, with an emphasis on directed message passing. By explicitly modeling the directionality of transactions, MP-GCN enables a multi-hop aggregation mechanism that extends beyond immediate (first-order) neighbors to incorporate information from more distant nodes in the transaction graph. This design carefully integrates features along the flow of transactions, allowing the model better to capture structural and behavioral patterns characteristic of phishing activities. Experimental results demonstrate strong classification performance, highlighting the critical role of directionality in distinguishing phishing from legitimate addresses.

- **Standard GNN Architectures & Autoencoders:** Another line of research applies GNNs with relatively minimal graph or feature engineering. Reference [128] develop a pipeline that combines random-walk embeddings for Ethereum role classification, such as identifying exchanges or miners with a GCN layer for final predictions. for final predictions. Integrating random-walk embeddings with GNN-based feature aggregation demonstrates robust performance across large-scale label sets. Reference [129] enhance suspicious address detection on Bitcoin by introducing moment-based features, including the variance and skewness of transaction amounts, into a lightweight GCN architecture, achieving both faster convergence and strong detection accuracy. Reference [130] frame Ethereum anomaly detection as a one-class classification task, employing a GNN-based autoencoder to learn node representations from transaction graphs and identify anomalies based on reconstruction error. In this setting, the autoencoder is trained exclusively on benign transaction data, enabling the detection of anomalous behavior as deviations from learned normal patterns. This method outperforms conventional anomaly detection approaches such as Isolation Forest and SVM, particularly under conditions of severe class imbalance. Finally, [131] apply standard GCN and GAT to anti-money laundering and counter-financing of terrorism (AML/CFT) detection on Bitcoin transaction networks. While GAT yields a modest performance improvement over GCN, both architectures offer substantial gains relative to simpler graph-based heuristics.
- **Domain-Specific & Novel Neural Architectures:** Other works devise more domain-specific neural network architectures tailored to unique aspects of cryptoasset data. Reference [132] transform Ethereum bytecode and Application Binary Interface (ABI) data into grayscale images and employ an attention capsule network for Ponzi scheme detection. This architecture integrates capsule networks that preserve hierarchical spatial relationships in data with an attention mechanism that selectively emphasizes salient features. The resulting attention-augmented capsules effectively capture code-level patterns in visual representations, achieving an F1 score of approximately 98.38%. Reference [133] introduce ChaosNet, a biologically inspired artificial neural network that emulates chaotic dynamics observed in biological neurons using chaotic neuron models based on Generalized Lurõth Series (GLS) maps. Applied to Ethereum address classification, the model demonstrates strong generalization and maintains competitive or superior accuracy with fewer training samples. Meanwhile, [134] diverges from transaction-level GNNs and applies a standard feedforward NN to identify a day-of-the-week effect in cryptoasset pricing. Although not focused on GNNs or anomaly detection, the study illustrates how deep learning architectures can uncover subtle cyclical patterns in crypto market behavior. Reference [135] propose a random-paced structure-to-vector embedding technique for user addresses in NFT and Ethereum networks. This method captures multi-scale structural identities—encompassing local connectivity, community-level relationships, and global structural roles—by sampling structural information at varying temporal or topological "paces." The resulting embeddings support high classification accuracy in detecting malicious nodes within metaverse-based financial environments. Finally, Hu et al. [136] introduce SCSGuard, which adopts a contract-level perspective by mapping Ethereum bytecode into opcode sequences and detecting scams using a Gated Recurrent Unit (GRU) network. GRUs, a type of recurrent neural network, are particularly effective at capturing temporal dependencies in sequential data. Combined with an attention mechanism, the model achieves strong performance in identifying Ponzi and Honeypot contracts by learning critical opcode patterns indicative of fraudulent behavior.

Despite their notable achievements, the methods in 10, spanning temporal GCNs, Transformer-based models, heterogeneous graph networks, and specialized neural architectures, exhibit both strengths and challenges. On the positive side, time-aware GNNs and Transformer hybrids excel at capturing dynamic or long-range dependencies, allowing the detection of subtle shifts in transaction patterns that simpler baselines would miss. Heterogeneous or multi-channel GNNs can handle different transaction types, e.g., contract calls versus standard transfers, improving expressiveness when dealing with complex blockchain ecosystems. Further, focusing on local subgraphs or star-shaped neighborhoods often offers computational efficiency, making it feasible to classify

TABLE 10. **Deep learning & Graph neural networks.**

Paper	Techniques	Blockchain	Data source	Anomaly	Measure
[115]	GCN, LSTM	BTC	On-chain (transaction)	money laundering	accuracy:0.97
[116]	GCN	ETH	On-chain (transaction)	money laundering	accuracy:0.97 recall:0.85 F1:0.89
[117]	GCN	ETH	On-chain (transaction)	malicious trading	precision:0.88 recall:0.93 F1:0.93
[118]	Skip-Gram (deep learning)	BTC	On-chain (transaction)	forecasting BTC transactions	accuracy:0.7
[119]	transformer (deep learning)	ETH	On-chain (transaction)	phishing attacks, de-anonymization	precision:0.7 recall:0.6 F1:0.65
[120]	transformer (deep learning), VAE	ETH (Olympus DAO)	On-chain (transaction)	fraudulent	N/A
[121]	GNN, RGCN, GraphSAGE, GAT	ETH	On-chain (transaction)	phishing fraud	precision:0.89 recall:0.74 F1:0.81
[122]	AMBGAT (GNN)	ETH	On-chain (transaction)	fraudulent of IoT	recall:0.85 F1:0.87
[123]	HGATE (GNN)	ETH	On-chain (transaction)	fraudulent	F1:0.97
[124]	HGATE (GNN)	BTC	On-chain (transaction)	fraudulent	F1:0.78
[125]	GNN	ETH	On-chain (transaction)	phishing scams	precision:0.81 recall:0.82 F1:0.82
[126]	MLP	ETH	On-chain (transaction)	phishing scams	precision:0.98 recall:0.99 F1:0.97
[127]	GCN	ETH	On-chain (transaction)	phishing nodes	precision:0.93 recall:0.90 F1:0.92
[128]	LSTM, GCN	ETH	On-chain (transaction)	phishing scams	AUC:0.93 F1:0.81
[129]	GCN	BTC	On-chain (transaction)	user behaviour	F1:0.43 recall:0.48
[130]	OCGNN (GNN)	ETH	On-chain (transaction)	user behaviour, smart contract-related attacks	accuracy:0.86 recall:0.82
[131]	GCN, GAT	BTC	On-chain (transaction)	AML/CFT	recall:0.79 F1:0.84
[132]	attention capsule network (deep learning)	ETH	On-chain (transaction)	ponzi scheme	F1:0.98 F1:0.84
[133]	ChaosNet (deep learning)	ETH	On-chain (transaction)	fraudulent	F1:0.79
[134]	feedforwad-NN	BTC,ETH, ADA	Off-chain (closing prices)	day-of-the-week anomaly	MSE:0.0009 - 0.0026
[135]	GNN, SVRP	ETH (NFT, metaverse)	On-chain (transaction)	fraudulent	accuracy:0.99 F1:0.96
[136]	GRU (recurrent NN)	ETH	On-chain (transaction)	ponzi schemes, honeypot contracts	accuracy:0.94 F1:0.96 recall:0.98

suspicious addresses on large-scale graphs. Nonetheless, several limitations remain. Many of these frameworks require extensive label availability or rely on carefully tuned hyperparameters for performance; label scarcity and class imbalance hamper generalization. Domain-specific approaches like those processing raw bytecode or capturing chaotic neuron behaviors can be challenging to extend across multiple blockchain platforms with differing transaction structures. In addition, interpretability remains a challenge; while attention mechanisms partially address transparency, fully justifying why specific nodes or edges drive the classification often requires additional heuristics. Finally, real-time deployment in fast-paced contexts such as DeFi demands further work on scalability and latency. These gaps suggest avenues for future research, such as self-supervised or active-learning strategies for label-constrained scenarios, multi-chain or cross-chain anomaly detection architectures, and better interpretability frameworks to align with regulatory requirements.

More broadly, these ML-based anomaly detection strategies face several interrelated limitations and open research directions that warrant further exploration. First, most supervised methods require large, high-quality labeled datasets, which can be impractical due to data scarcity, evolving fraud tactics, and class imbalance, where legitimate ones dwarf fraudulent samples. This imbalance forces difficult trade-offs between metrics such as F1, recall, and accuracy, requiring careful calibration or advanced oversampling/undersampling. Second, interpretability remains a critical hurdle; ensemble and deep architectures often act as black boxes, making it challenging to explain why transactions or addresses are flagged as anomalous. Third, most studies focus on single blockchain ecosystems; future research could expand to multi-chain or cross-chain detection, given that malicious activities often spread across platforms. Fourth, real-time detection poses an additional challenge in dynamic environments such as DeFi, demanding low-latency, scalable methods that can handle continuous streams of transactions. Finally, an ensemble-driven paradigm where multiple diverse models such as RF, GNN, and Transformers are simultaneously trained and stacked represents a promising avenue for boosting robustness and generalization, especially under adversarial conditions. Exploring these directions, particularly self-supervised, active-learning approaches for label-scarce scenarios and improved interpretability frameworks, would further advance the reliability and practical deployment of ML-based solutions in the cryptoasset domain.

D. HEURISTIC-BASED

Heuristic-based anomaly detection methods utilize expert-driven or rule-based models to identify anomalous or fraudulent patterns within cryptoasset transaction networks. These methods range from forensic and analytical modeling to specialized protocol designs and cryptographic techniques. In contrast to statistical or machine learning-based techniques, heuristic approaches often incorporate domain-specific knowledge, emphasizing interpretability, transparency, and regulatory compliance.

1) FORENSIC & ANALYTICAL MODELING

Forensic and analytical modeling approaches rely on expert-defined heuristics and empirical observations to trace suspicious activities, particularly those related to money laundering, ransomware payments, and market manipulation. Summary of the studies categorized under this category is presented in table 11.

- **Modeling & Analysis of Mixing Operations:** A significant focus lies on understanding and modeling cryptoasset mixing services used for laundering. A heuristic-based goal modeling framework was introduced to detect and categorize roles involved in Bitcoin money laundering activities, particularly in mixing operations [137]. A mixing operation refers to a process where illicitly obtained cryptoasset is combined with funds from other sources, using numerous intermediate addresses, to obscure its original source and destination, thereby complicating tracking efforts. The approach classifies Bitcoin addresses involved in these activities into three distinct roles based on their transaction behaviors and structural patterns: entry addresses (communicators), which initially receive illicit funds, kernel addresses (soldiers), intermediary addresses frequently used to obscure and redistribute funds within the mixing network, and exit addresses (communicators), where funds ultimately leave the network, typically toward fiat gateways or cryptoasset exchanges. By heuristically modeling these roles through transaction characteristics, timing patterns, and relational structures, the method systematically uncovers laundering activities within complex Bitcoin transaction graphs.

Further analysis of mixing operations provides a more detailed understanding of the methods used to obscure illicit activity within blockchain networks. Modern services such as MixTum, Blender, and CryptoMixer employ advanced techniques, including randomized transaction delays, multiple recipient addresses, partitioning transfers into smaller amounts, and the use of "sweeper" transactions to periodically consolidate dispersed funds before redistribution [138]. Temporal and structural features, such as deposit–withdrawal intervals and address reuse patterns, exhibit consistent behaviors. The analysis emphasizes "chain-level" patterns, focusing on sequences of transactions rather than individual ones. Patterns such as short inter-transaction intervals, repeated fund-splitting, and systematic address reuse are commonly observed. By tracing how outputs from one transaction serve as inputs to the next and identifying recurring features, such as typical transaction sizes or timing intervals, it is possible to detect mixer-related transaction chains with greater confidence.

A complementary abstraction model has been proposed to analyze both centralized and decentralized mixers [139]. This three-phase model includes: taking inputs, performing the mix, and sending outputs. Transaction-level analysis of platforms such as ChipMixer, Wasabi Wallet, and ShapeShift demonstrates how asset-swapping mechanisms and anonymity set construction obscure fund provenance. Two frequently observed techniques are *peeling chains*, where small outputs are incrementally extracted over sequential transactions, and *obfuscating mechanisms*, where transactions are aggregated into anonymity sets to disrupt linkage analysis. While these techniques are intended to hinder tracking, they leave behind identifiable traces that can be systematically analyzed.

- **Context-Aware Taint Analysis:** Improvements to traditional taint analysis have also been introduced to enhance the precision of tracing illicit Bitcoin flows [140]. Taint analysis marks coins as "tainted" when they are linked to illegal activity and follows their movements through the blockchain. Rather than tracking every transaction indiscriminately, the refined approach incorporates address profiling—classifying addresses as exchanges, darknet markets, payment processors, gambling services, and other categories, to determine which paths are relevant. Two context-based strategies are introduced to adapt the analysis depending on the situation. Evaluation metrics based on expected behavior of illicit funds and observable blockchain patterns are also defined to assess accuracy. This context-aware method reduces unnecessary tracking and improves the detection of meaningful transaction trails.
- **Empirical Laundering Patterns:** Further investigations focus on how cybercriminals convert stolen Bitcoin into usable funds exceeding $11 million [141]. One case study analyzes the Conti ransomware operation, a prominent ransomware-as-a-service (RaaS) group active until 2022, which targeted businesses and critical infrastructure with high ransom demands [142], [143]. Findings show that while some actors employ advanced obfuscation, many rely on simpler methods such as repeated use of centralized exchanges, minimal layering, or peer-to-peer transfer networks. Even for high-value ransomware proceeds, laundering patterns often involve basic fund splitting and direct cash-out services, challenging the assumption that complex chains and multiple mixers are always used.

  Additional analysis has focused on fraud and scams in the decentralized finance (DeFi) ecosystem, particularly involving ERC-20 tokens on the Ethereum blockchain [144]. Using open-source investigative methods, including transaction tracing tools like Etherscan and smart contract analysis tools such as Slither, patterns of illicit behavior such as rug pulls, pump-and-dump schemes, and subsequent laundering activities have been identified. These techniques allow for examination of transaction histories, token flows, smart contract behavior, and bridging activities across chains. Malicious actors often attract victims through decentralized exchanges, extract funds, and then move proceeds through mixers or cross-chain bridges. While the technical complexity of the DeFi infrastructure suggests the potential for sophisticated laundering, findings indicate that many schemes rely on relatively simple methods, such as cashing out via centralized exchanges or using basic bridging strategies. These actions leave identifiable on-chain traces that can be systematically analyzed to uncover fraud patterns and actor linkages.

2) PROTOCOL & CRYPTOGRAPHIC DESIGN

Protocol and cryptographic design approaches focus on embedding security features within blockchain systems or evaluating existing mechanisms to detect weaknesses. Instead of concentrating exclusively on user-level transaction flows, these studies often scrutinize the underlying consensus protocols, deposit frameworks, and oracle implementations to

TABLE 11. Forensic & Analytical modeling.

Paper	Blockchain	Data source	Anomaly	Measure
[137]	BTC	On-chain (transaction)	money laundering (mixing operation)	N/A
[138]	BTC	On-chain (transaction), Off-chain (mixing address) (BitcoinTalk & Reddit)	money laundering (mixing operation)	recall: 1.0
[139]	BTC	On-chain (transaction)	money laundering (mixing operation)	recall: 0.92
[140]	BTC	On-chain (transaction)	money laundering, cryptoasset theft	N/A
[141]	BTC	On-chain (transaction)	money laundering related to Conti	N/A
[144]	ETH (ERC-20)	On-chain (transaction)	rug pulls, pump-and-dump	N/A

TABLE 12. Protocol & Cryptographic design.

Paper	Blockchain	Data source	Anomaly	Measure
[145]	protocol level	BFT protocols and message structures	Byzantine faults	N/A
[146]	BTC	On-chain (transaction)	anti-fraud	N/A
[147]	ETH and related tokens	On-chain (transaction)	design DeFi (oracle outages)	N/A

ensure robust cryptographic guarantees and resilience against malicious actors. For a summary of the studies discussed in this category, refer to table 12

- **BFT Protocol Forensics & Accountability:** One line of work investigates Byzantine Fault Tolerance (BFT) protocol forensics, which formalizes post-violation diagnostics and accountability in consensus protocols [145]. When safety violations occur, e.g when more than a threshold number of nodes act maliciously, the protocol is expected to generate cryptographically verifiable evidence that identifies the responsible replicas. To capture a protocol's forensic capabilities, its support is summarized by a triplet (m, k, d) where m the maximum number of malicious nodes under which the protocol can still provide forensic evidence, k the minimum number of honest nodes' transcripts required to reliably prove culpability and d the number of Byzantine nodes that can be held accountable after an agreement violation. Analysis of protocols such as PBFT, HotStuff, VABA, and Algorand shows that even minor design variations can significantly affect these forensic parameters. For example, under certain configurations, e.g. PBFT-MAC, HotStuff-null, and Algorand, even if transcripts from all honest nodes are available, no meaningful forensic evidence is produced, $d = 0$. By examining message structures and quorum certificates, the study outlines conditions under which sufficient forensic data can be collected to reliably identify adversarial nodes. This systematic approach enhances the ability of BFT systems to recover from faults and strengthens their defense against coordinated attacks.
- **Cryptographic Endorsement for Secure Deposits:** Another line of research explores how to secure Bitcoin-based deposits in specialized environments, such as connected vehicles, automobiles equipped with internet connectivity allowing them to communicate with other devices both inside and outside the vehicle, enabling various applications from navigation and infotainment to advanced driver assistance systems and vehicle-to-vehicle communication. A novel "Bitcoin-to-Connected-Vehicle" (Bit2CV) scheme which uses cryptographic endorsements to verify the origins of deposited funds has been proposed [146]. In this scheme, the anti-fraud measures are primarily based on a cryptographic endorsement procedure that leverages threshold signatures $\sigma = (\sigma_{agg}, \varepsilon)$, where σ_{agg} is an aggregated signature and ε represents a set of indices corresponding to the signers who participated in creating the signature. In this scheme, a vehicle must collect endorsements from a threshold number of authorized parties to verify the origin of deposited funds, thereby providing robust anti-fraud measures while maintaining compatibility with existing Bitcoin infrastructure.
- **DeFi Oracle Security & Design:** Finally, a broader examination of decentralized finance oracles [147] focuses on how blockchain protocols acquire and validate real-world data, particularly market prices and exchange rates, without relying on a single trusted party. The study investigates mainstream DeFi platforms built primarily on Ethereum, which commonly involve cryptoassets such as ETH, DAI, MKR, AMPL, and SNX. In these systems, a small set of whitelisted oracles provides data that is aggregated to determine on-chain prices, making the system's integrity highly dependent on a few key actors. Analysis of real-world oracle deployments shows that reported prices often deviate from current exchange rates, and oracles can suffer from operational issues and anomalies. A comparison of designs, including those used by MakerDAO (DAI and MKR), AmpleForth (AMPL), and Synthetix (SNX), reveals that each employs unique mechanisms for data aggregation and validation. Proposed improvements include stronger cryptographic binding of data, more transparent governance over oracle selection and operation, and robust mechanisms for detecting and mitigating anomalous data, ensuring that on-chain protocols accurately reflect off-chain reality.

3) HEURISTICS FOR SECOND-LAYER & ON-CHAIN EXPLOITS

Second-layer networks, such as the Lightning Network (LN), enable off-chain transactions and micro-payments to improve blockchain scalability, but also introduce new vectors for misbehavior. The Lightning Network operates by creating payment channels between users, allowing them to conduct multiple transactions off the main Bitcoin blockchain. Only the opening and closing of these channels are recorded on-chain. However, this opacity also presents challenges for monitoring and security. Table 13 summarizes the studies covered in this category.

- **Lightning Network Analysis:** Research in this area focuses both on identifying LN activity from on-chain data and analyzing vulnerabilities within the LN protocol itself. One study [148] evaluates multiple heuristics to identify these LN-related transactions within the on-chain data. This research explores what can be deduced and inferred about the layer-two overlay network based on the transactions recorded in the ledger. The analysis shows that over 75% of all 2-of-2 multisignature (2of2 multisig) transactions on the Bitcoin using Pay-to-Witness-Script-Hash (P2WSH) can be linked to LN channels. By correlating observable patterns, e.g. channel opening and closing, with known LN addresses, the study demonstrates that it is possible to infer aspects of off-chain activity from on-chain records, even if only part of the LN topology is revealed.
  Complementary work [149] investigates routing vulnerabilities in the LN. The findings indicate that adversaries can strategically deploy LN channels with artificially low fees to attract payment routes, effectively hijacking the network's routing topology. This tactic allows them to exert undue influence, potentially censoring or delaying transactions. The study reveals a fundamental tradeoff: rational LN nodes, seeking efficient (low-fee) routes, become susceptible to exploitation. To mitigate this risk and enhance security, nodes must incur higher transaction fees to avoid predictable routing patterns. The study reveals that routing in LN is highly centralized: nearly 60% of all routes pass through only five nodes, and 80% through just ten nodes. This concentration exposes the network to denial-of-service attacks from a small set of colluding entities. Furthermore, the research models an external attacker establishing new LN links with minimal fees. Results indicate that creating as few as five such links can divert a majority 65%-75% of network traffic, regardless of the specific LN implementation. The cost of deploying these attack links is demonstrably low, underscoring the economic feasibility of routing-based exploits in the LN.
- **On-Chain Market Manipulation:** On the on-chain side, sophisticated manipulations occur on decentralized exchanges (DEXs) that rely on transparent smart contracts for trading. High-frequency or so-called "sandwich" attacks on Automated Market Maker (AMM) platforms such as Uniswap is studied [150]. In a sandwich attack, an adversary exploits the latency between transaction broadcast and execution by observing a pending transaction in the mempool, placing a buy order immediately before the victim's order (front-running), and then executing a sell order immediately afterward (back-running) to profit from the induced price movement. The study formalizes the attack mathematically using the constant product formula, which governs the price impact of trades on AMMs $x \cdot y = k$ where x and y represent the reserves of the two tokens in the liquidity pool, and k is a constant. Empirical evaluation shows that a single attacker can achieve an average daily revenue of approximately \$3,414 on Uniswap. These findings highlight that while the transparency of blockchain transactions enables verification and auditability, it also creates vulnerabilities that can be exploited for market manipulation, underscoring the need for improved safeguards in decentralized trading systems.
  In addition, a hybrid detection approach has been proposed to identify pump-and-dump (P&D) schemes on cryptoasset markets [151]. This method combines distance- and density-based anomaly metrics to detect sudden, suspicious price–volume movements across multiple exchanges. It reformulates the problem of contextual anomaly detection in time series data into a point anomaly detection problem by dividing the time series into frames, concatenating the data within each frame into high-dimensional data points, and projecting these points into a two-dimensional space using Principal Component Analysis (PCA). In this reduced space, established distance- and density-based techniques are applied to effectively detect anomalies. The approach consistently outperforms single-metric methods by capturing anomalous patterns that might be overlooked when using solely distance-based or density-based measures, resulting in a higher detection rate of P&D events across top-ranked exchange pairs and a lower rate of false positives overall.
  At a broader scale, an agent-based study simulates price manipulation in the Bitcoin market driven by Tether injections [152]. The simulation models both typical trader behavior and a fraudulent agent that repeatedly injects Tether on selected exchanges and makes sustained Bitcoin purchases. In markets with thin liquidity, these purchases push prices upward, attracting additional momentum-following traders and magnifying the effect. The malicious agent then strategically sells small volumes of Bitcoin to recoup funds and satisfy "proof of capital" requirements, typically aligned with end-of-month reporting. The results demonstrate that this feedback loop of Tether inflows and controlled Bitcoin sell-offs can trigger large price swings in an illiquid market. The study concludes that concentrated control over stablecoin issuance, combined with limited liquidity, leaves the Bitcoin ecosystem vulnerable to manipulation by a single actor. It also suggests that more frequent audits of stablecoins and efforts to deepen market liquidity could help reduce the risk of such price inflation schemes.

Heuristic-based anomaly detection methods for cryptoasset transactions excel in interpretability and domain

TABLE 13. Heuristic-based methods.

Paper	Blockchain	Data source	Anomaly	Measure
[148]	BTC	On-chain (transaction)	lightning network	recall: 0.9, precision:0.99
[149]	BTC	On-chain (transaction)	lightning network	N/A
[150]	ETH (Uniswap)	On-chain (transaction)	sandwich attack	N/A
[151]	BTC	On-chain (transaction)	pump-and-dump	N/A
[152]	BTC	On-chain (transaction), Off-chain (order-book, price/volume)	price manipulation	N/A

specificity, allowing investigators to quickly flag suspicious behaviors (e.g., short inter-transaction intervals, mixing, or protocol exploits) without requiring a large training dataset. Such heuristics are relatively straightforward to implement, rely on known illicit patterns, and can be tuned to specific network features (like repeated fund-splitting or cross-chain bridging). However, they may fail to detect complex or evolving laundering strategies beyond the scope of pre-defined rules, leading to higher false negatives as criminals adapt. In addition, purely heuristic approaches can be overly rigid, generating potential false positives whenever normal users share superficial similarities with illicit addresses (e.g., frequent transactions). Nonetheless, when integrated into a broader detection pipeline, potentially employing machine learning, address classification, and external intelligence feeds, heuristic triggers can act as the "first line of defense," rapidly filtering out large volumes of routine activity while flagging suspicious outliers for deeper investigation. This synergy between domain-driven heuristics and automated analysis tools thus offers promising avenues for new research, such as refining heuristics to detect novel off-chain exploits or designing feedback loops that update detection rules based on confirmed threat actor behaviors.

IV. CHALLENGES, LIMITATIONS, AND FUTURE RESEARCH DIRECTIONS

This SoK has reviewed a collection of 103 papers centered on anomaly detection within cryptoasset ecosystems, classifying the employed techniques into four primary categories: statistical analysis, network analysis, machine learning, and heuristic-based methods, as detailed in Section III. This section synthesizes these findings, providing a comparative analysis across these categories. It also identifies significant overarching challenges prevalent in the field and delineates promising future research trajectories intended to guide subsequent investigations in this dynamic domain.

A. COMPARATIVE ANALYSIS OF DETECTION CATEGORIES

After evaluating the four classes of methodology, we found distinct characteristics and trade-offs concerning their data requirements, detection capabilities, interpretability, and robustness. Table 14 provides a synthesized comparison of these methodologies, summarizing their applications, performance, and qualitative features based on the 103 studies reviewed. While a direct comparison of performance metrics is challenging due to the lack of standardized benchmark datasets across studies, the table reveals clear patterns that highlight the strengths and weaknesses inherent to each approach.

Regarding data requirements and underlying assumptions, Statistical methods commonly assume specific data distributions, potentially limiting their effectiveness in volatile cryptoasset markets, as they typically analyze numeric transaction metrics rather than the underlying network structure. Network Analysis techniques primarily leverage the transaction graph's topology, making them less reliant on distributional assumptions but sensitive to how the graph is constructed and computationally intensive for large networks. Machine Learning approaches vary substantially based on their subtype: supervised methods depend heavily on labeled datasets, which are often scarce; unsupervised and deep learning models circumvent labeling limitations but necessitate extensive datasets and careful feature engineering. Heuristic methods differ from the others by relying on explicitly encoded domain expertise rather than extensive data. Although data-light, these methods require continual expert input to define and update their rules.

Regarding detection capabilities, Statistical methods are particularly effective at identifying point anomalies, such as sudden numerical deviations in transaction metrics. Network Analysis excels at detecting structural and collective anomalies, like coordinated fraudulent activities or network attacks that leave distinct topological traces. Machine Learning methods offer broad capabilities, identifying not only point anomalies but also complex contextual and collective patterns, even uncovering previously unseen threats through learned models. Heuristic approaches are highly effective for addressing well-known vulnerabilities or explicit anomalous patterns, such as specific smart contract exploits, by directly encoding domain-specific knowledge into rules.

Interpretability varies significantly across methodologies. Statistical methods and Heuristic rules typically offer high interpretability due to their transparent logic and straightforward analytical frameworks. Network Analysis provides moderate interpretability, enabling visual representations of detected graph structures; however, advanced network metrics may be less intuitive to interpret. Machine Learning methods range widely, with simpler algorithms offering clear insights into their decision-making processes. In contrast, complex models, particularly deep neural networks, often operate as "black boxes," posing challenges for forensic analysis and trust despite their powerful analytical capabilities.

Scalability and computational requirements introduce additional trade-offs. Statistical and Heuristic methods usually have low computational demands, making them suitable for real-time anomaly detection. Conversely, Network Analysis methods can become computationally intensive

TABLE 14. Comparison of anomaly detection methodologies for crypto assets (numbers in parentheses indicate the number of studies involved).

Methodology	Crypto assets	Anomaly Studied	Key Performance Metric	Data source	Strengths & Practical Insights	Limitations
Statistical Analysis	BTC (9), ETH (6), LTC (2), BCH (2), MONA (1), XRP (1)	Fraud (3) MM & Price (3) Consensus Attacks (3) User Behaviour (1)	statistical significance tests F1: 0.93 statistical significance tests Granger causality	On-chain + Off-chain Off-chain On-chain On-chain	• Highly interpretable • Computationally efficient for simpler methods; suitable for initial screening • Effective for detecting point anomalies, such as sudden numerical deviations in transaction values or prices	• Sensitive to data distributions and can be less effective in volatile crypto markets where "normal" behavior constantly changes • Advanced models are computationally expensive and may not be suitable for real-time detection.
Network Analysis	BTC (19), ETH (10), EOSIO (2), XRP (1)	User Behaviour (18) Fraud (6) MM & Price (4) Consensus Attacks (1)	N/A N/A N/A N/A	On-chain + Off-chain On-chain On-chain Off-chain	• Excellent for detecting structural and collective anomalies like coordinated fraud, money laundering rings, or centralized hubs • Moderate interpretability via graph visualizations • Can capture bursty, short-lived behaviors that static methods might miss.	• High computational cost and faces significant scalability challenges as blockchain networks grow. • Performance is sensitive to graph construction choices, e.g., time window and node definition.
Supervised, Unsupervised and Semi-supervised Learning	BTC (16), ETH (12)	Fraud (15) MM (6) User Behaviour (5) Consensus Attacks (1)	AUC: 0.99, Accuracy: 0.99, F1: 0.96 Accuracy: 0.99, AUC: 0.99 Recall: 0.94 Accuracy: 0.90	On-chain On-chain On-chain + Off-chain On-chain	• Supervised models achieve high accuracy when quality labels are available • Unsupervised models excel at finding novel anomalies and work well in label-scarce environments • Highly flexible for detecting a wide range of anomaly types	• Supervised methods are heavily dependent on scarce, high-quality labels and suffer from class imbalance. • Unsupervised methods can have high false-positive rates and limited interpretability
Deep Learning & Graph Neural Networks	ETH (17), BTC (6), ADA (1)	Fraud (16) MM & price (3) User Behaviour (3)	Recall: 0.99, F1: 0.97, Accuracy: 0.97 F1: 0.98, Recall: 0.98 MSE: 0.0009 - 0.0026 Accuracy: 0.86, F1: 0.93	On-chain On-chain + Off-chain On-chain	• Automatically learn complex patterns from raw data, reducing the need for manual feature engineering • Capturing dynamic, temporal, and relational dependencies within transaction graphs • Highly effective for subgraph-level analysis, making them ideal for tasks like phishing detection	• Often operate as "black boxes," making results difficult to interpret • Real-time deployment can be limited by model latency and scalability challenges
Heuristic-Based	BTC (10), ETH (3), Protocol (1)	Fraud (5) Protocol & Second-Layer Exploits (4) MM & price (3) Consensus Attacks (2)	N/A N/A N/A Precision: 0.99, Recall: 0.90	On-chain + Off-chain On-chain + Protocol-level On-chain + Off-chain On-chain	• Highly interpretable as the detection rules are explicitly defined by experts • Very effective for detecting known attack patterns and well-defined vulnerabilities	• Brittle and ineffective against novel or evolving threats that do not match predefined rules • Requires continuous manual updates by domain experts to remain effective as attacker tactics change • Can generate false positives if normal behavior superficially matches a rule

due to the complexity of processing large-scale blockchain transaction graphs. Machine Learning methods, particularly deep learning approaches, require significant computational resources during model training, although inference can be relatively fast and scalable once trained.

Finally, adaptability and robustness highlight further differences. Statistical methods often face challenges in environments with rapid concept drift, which is common in cryptoasset markets and requires frequent recalibration. Network Analysis methods show robustness against certain noise and small-scale manipulations but remain sensitive to substantial topological changes or sophisticated adversarial attacks. Machine Learning approaches can adapt through retraining yet remain susceptible to adversarial attacks specifically designed to evade detection. Expanding the research horizon further requires integrating cross-disciplinary paradigms, such as causal inference to distinguish intentional manipulation from actual market volatility, and reinforcement learning for dynamic monitoring of user behaviors. Additionally, advancing privacy-preserving analytics through techniques like zero-knowledge proofs and cross-chain graph alignment will be pivotal for maintaining regulatory compliance without compromising user data privacy. Despite their interpretability, heuristic methods are brittle and effective for known threats but limited in their ability to generalize and necessitate ongoing manual updates to maintain detection effectiveness.

The comparative analysis presented in Table 14 underscores that no single methodology is universally superior. The optimal choice is context-dependent, balancing the need for high performance against the practical constraints of data availability, the demand for robustness against novel threats, and the requirement for interpretability. From a practical standpoint, these trade-offs suggest a stratified deployment strategy, i.e. heuristic rules serve as an interpretable first line of defense for known threats, while unsupervised learning is essential for spotting novel patterns in label-scarce environments. Conversely, deep learning workflows are best reserved for high-volume, historical analysis where computational resources and labeled data are sufficient to support complex model training.

B. REAL-WORLD ANOMALIES

Synthesizing the findings from Section III, this section connects the surveyed methodologies to their application in detecting prominent real-world anomalies. By grounding the taxonomic analysis in concrete use cases, we can better evaluate the strengths and limitations of current techniques and highlight where certain methods are most effective. The following discussion focuses on several key real-world anomalies, evaluating how the surveyed methodologies have been applied in practice to detect them.

1) MARKET MANIPULATION AND PRICE-RELATED ANOMALIES

P&D and price–trend manipulation are typically executed via coordinated bursts in price–volume and bursts in trading activity. One effective approach uses signature methods [43] to transform raw trade data—price, volume, side, and timestamp into powerful features. This technique can detect P&D with F1 score up to 88%, making it highly competitive with supervised methods while relying only on publicly available trade histories. Similarly, forecasting-anomaly pipelines use models like SARIMAX to flag periods where price trends deviate significantly from predictions. The highest-volume accounts active during these anomalous windows are then flagged as potential manipulators. This approach is highly successful, achieving an F1 score of up to 93%. For DeFi-specific scams like rug pulls, which often combine P&D tactics, forensic investigation using open-source tools like Etherscan and Slither can reconstruct the entire scam lifecycle [144]. This method reveals the common pattern of token creation, liquidity seeding, orchestrated buys, and eventual liquidity removal. This analysis also shows that the subsequent money laundering methods are often unsophisticated. Finally, a hybrid distance-density framework improves detection by reducing the dimensionality of price-volume data with PCA [151]. This allows a combination of distance- and density-based outlier scores to identify abrupt trading surges with a lower false-positive rate than single-metric methods.

On the other hand, price-related anomalies were studied by treating unusual price co-movements as market-level anomalies rather than explicit manipulation. A network-centric line links transaction-network structure to price e.g. principal-component dynamics of Bitcoin's address network correlate with market regimes [70], and weekly correlation-tensor/PCA snapshots of XRP transaction networks produce singular-value signals that align with subsequent price bursts [76]. A modeling line uses agent-based simulations to show how concentrated stable-coin inflows (e.g., Tether) in thin liquidity can amplify price swings consistent with manipulation-driven bubbles and drawdowns [152].

These studies show that market manipulation can be detected through two complementary lenses: direct analysis of market data where statistical and machine learning models identify anomalous price-volume patterns in real-time or historically and indirect analysis of underlying market structure, where changes in transaction network topology or simulated agent behaviors signal price instability and manipulation risk.

2) EXCHANGE EXPLOIT: THE Mt.Gox CASE

As the dominant Bitcoin exchange until its 2014 collapse, Mt. Gox is a central case study for exchange-level manipulation. Analyses of transaction history between 2011 to 2013 reveal that accounts trading at extreme, unrealistic prices formed dense clusters and unusual motifs (triangles, self-loops). Temporal SVD showed these abnormal accounts were tightly correlated with Bitcoin price movements, consistent with liquidity creation and fake volume [19]. A complementary approach models monthly transaction networks with hidden Markov tensor methods and monitors latent variables using MEWMA control charts. This framework flags the late-2013 period as ''out-of-control,'' providing statistical evidence of manipulation without requiring explicit labeling [52]. Broader network studies confirm Mt. Gox's systemic role: structural break analysis shows that after its bankruptcy, heavy-tailed out-degree distributions lost stability, and network heterogeneity lost predictive regularity for price. This indicates that Mt. Gox acted as a central hub driving both liquidity and volatility [74].

Together, these methods i.e. graph classification with SVD, latent-variable monitoring, and structural break analysis—highlight how different anomaly detection frameworks can reconstruct and quantify the manipulation that contributed to Mt. Gox's downfall.

3) MONEY LAUNDERING & TERRORIST FINANCING

The detection of illicit financial flows is approached by analyzing on-chain data in relation to real-world events and network behavior. One line of research focuses on terrorist financing [54] builds a labeled map of large on-chain service providers (exchanges, mixers, gambling, mining, dark markets) and then monitor for abnormal transfer volume around major terrorist attacks. This approach identifies significant increases in funds flowing into unregulated exchanges and mixers, the channels used to move funds from organizers to local operatives and to launder them before cash-out. Forensic accounting on specific events, such as the Sri Lanka Easter bombing, corroborates these findings and helps build machine learning models that use on-chain flow features for risk prediction. Practically, a cross-asset move like BTC to XRP shows up as funds leaving Bitcoin into known exchange clusters, so the detector keys on the inflow/outflow bursts to those exchange wallets rather than the off-chain conversion step.

In more general case of money laundering, various machine learning models are applied. A case study of the Upbit hack [84] on Ethereum characterizes ML networks by traditional traits such as fast-in/fast-out transfers and dense transaction clusters, providing concrete features for on-chain detection. On the Bitcoin network [91] (Elliptic dataset), graph embeddings are particularly effective, achieves approximately 92% accuracy, though performance can degrade during market disruptions like dark-market shutdowns. The performance of these models can be improved with specialized features [92], [103]. More advanced architectures like a temporal-GCN [115] and LB-GLAT [116] explicitly model transaction sequences and graph directionality to address challenges like over-smoothing, achieving high accuracy and F1-scores.

A specific challenge within ML is detecting mixing services, which are purpose-built to obfuscate fund origins. Research in this area often interacts with mixers to obtain ground-truth data, which is then used to identify transaction- and chain-level patterns, e.g., I/O structure, sweeper transactions [137]. Mixer mechanisms are formalizes as either swapping (using peeling chains) or obfuscating (using CoinJoin), with heuristics identifying over 92% of obfuscating transactions [138]. To improve tracing, context-aware taint analysis [139] uses address profiling to define logical exit points (e.g., exchanges, gambling sites), pruning irrelevant transaction paths. However, empirical studies show a contrast to these sophisticated tools, revealing that many criminals use surprisingly unsophisticated laundering methods, often preferring direct transfers to centralized exchanges [140], [141].

4) PONZI SCHEMES AND PHISHING

The detection of user-facing scams like Ponzi schemes and phishing relies heavily on machine learning, with distinct strategies tailored to each threat. For Ponzi schemes, research focuses on pre-deployment detection by analyzing the smart contract itself. One approach analyzes contract artifacts, such as mapping bytecode or Application Binary Interface (ABI) features into images for CNN and Capsule Network pipelines, which effectively learn patterns in the contract's logic and function calls [93], [95], [98], [132]. A complementary method uses attention-augmented RNNs to learn directly from n-grams of bytecode sequences, creating generalizable detectors for Ponzi and related scams [94], [96], [97], [136]. These studies show that a contract's static code footprint, including opcode frequency and control-flow structure, is highly discriminative for identifying Ponzi schemes even before any user transactions have occurred.

In contrast, phishing detection focuses on post-deployment analysis of transaction networks, where Graph Neural Networks (GNNs) are the dominant methodology. Early work established a baseline by creating transaction-aware network embeddings and applying one-class SVMs to handle the severe class imbalance between fraudulent and licit addresses [85], [90]. Current research builds on this with more advanced GNNs. Studies consistently find that heterogeneous GNNs, which explicitly model different node and edge types (e.g., EOA vs. contract, transfer vs. call), outperform simpler architectures [105], [107]. Other effective methods operate on a subgraph-level [119], [121], analyzing an address's local neighborhood with features like transaction value, gas, and time to achieve very high recall. The importance of time is also a key theme; models that create temporal edge embeddings [125], [126] or use pre-trained Transformers on raw transaction sequences report significant performance gains over static graph methods by capturing the behavioral rhythms of phishing attacks, such as fund consolidation and cash-out [127], [128].

The most effective strategy for Ponzi schemes is pre-deployment analysis of the contract's bytecode and ABI, as these static features provide accurate flags without needing on-chain history. For phishing, the best results come from combining graph structure with temporal data, using heterogeneous GNNs on transaction subgraphs enriched with time and value features, or employing sequence models like Transformers. In practice, a two-stage pipeline is effective: (1) pre-deployment screening for Ponzi-like bytecode patterns, followed by (2) post-deployment monitoring that fuses graph structure with temporal cues to detect phishing activity.

5) CONSENSUS LAYER ATTACKS

A primary concern is the 51% (or majority) attack, where a colluding group could rewrite transaction history. Empirical analysis of Bitcoin and Ethereum shows that mining power is increasingly concentrated among a small number of entities, challenging the assumption of decentralization and creating a tangible risk of a 51% attack [153], [154]. This makes continuous monitoring of miner shares and patterns in consecutive block production a critical early-warning system [57], [111].

Beyond direct majority control, more subtle strategic deviations like selfish mining (SM) also identified where miners selectively withhold newly found blocks to gain an advantage. Detection methods focus on the statistical anomalies this behavior creates, specifically in the frequency of consecutive block discoveries. One approach uses Miner Sequence Bootstrapping (MSB) [55], a simulation-based method, while a more direct statistical test uses the type II binomial distribution as a null model for honest mining [56]. These methods have identified statistically significant SM behavior, particularly in Monacoin and Bitcoin Cash.

For selfish-mining, miner and pair run-length tests with accurate miner attribution (clustering) are the most direct methods and have revealed real-world anomalies. For 51% attacks, continuous monitoring of pool shares and simple burst metrics provides actionable risk indicators, while generic one-class models offer a lightweight secondary screen.

C. MULTI-CHAIN ANOMALY DETECTION

While most anomaly-detection work is single-chain, a number of studies extend their analysis across multiple cryptoassets. These approaches, often comparative rather than fully integrated, reveal two crucial insights. First, that "normal" on-chain behavior is not uniform across blockchains and second, that illicit actors increasingly operate across these different ecosystems.

1) COMPARATIVE STRUCTURE & BEHAVIOR

Fundamental differences in network topology are evident across major blockchains. Monthly transaction networks for Bitcoin, Ethereum, and Namecoin all exhibit heavy-tailed degree distributions that deviate from simple power laws, while network statistics like degree assortativity reveal underscore structural differences between chains [16]. Building on this, multi-chain analyses of preferential attachment formalize how "rich-get-richer" dynamics drive hub formation in Bitcoin and Ethereum [73], while ERC-20 token networks often exhibit super-linear attachment, which accelerates the concentration of network activity into a few hubs [72]. This demonstrates that a detector calibrated to one chain's topology would likely fail on another, making multi-chain baselining essential for accurate detection.

2) MULTI-ASSET IRREGULARITIES

Methodologies that screen for macro-level irregularities are effective at flagging anomalous activity across multiple assets simultaneously. Robust distance metrics, such as Mahalanobis distances, can detect anomalies in return vectors across multiple cryptocurrencies simultaneously [45]. These results highlight periods like the 2021 "metaverse boom," where correlated surges flagged joint-market stress. Similarly, Benford's Law has been used to identify currencies whose transaction values deviate from expected statistical distributions. While Bitcoin and Ethereum conformed, others such as TENX, VERI, and DOGE showed anomalies linked to documented scandals [44]. Such macro-level methods serve as effective early-warning systems, flagging cross-asset irregularities that warrant deeper on-chain investigation.

3) MINING BEHAVIOR ACROSS PoW CHAINS

Mining-centric anomalies have been measured consistently across BTC, LTC, ETH, BCH, and MONA. The Miner Sequence Bootstrapping (MSB) model tests whether a miner appears too often in consecutive blocks relative to chance, flagging selfish strategies; a paired MSB extends to mining cartels [55]. A follow-on study generalizes the test and reports that Monacoin shows an unusually high fraction of abnormal miners, with persistent selfish-mining signals; Bitcoin Cash also exhibits bursts of abnormality, and cartel-like coordination is observed in MONA, ETH, BCH more than in BTC and LTC [56]. Monitoring miner-share concentration adds a complementary perspective and early-warning lens for 51% attack, with empirical miner-share profiles in BTC/ETH illustrating the practical value of such tracking [57].

4) SCALABLE SUPERVISED PIPELINES ACROSS CHAINS

On the supervised side, GPU-accelerated pipelines deploy SVM, Random Forest, and Logistic Regression on tens of millions of Bitcoin transactions and hundreds of thousands of Ethereum accounts, demonstrating that features + parallelism yield practical, near-real-time fraud detection across distinct networks [99].

5) CROSS-CHAIN ANOMALIES

Most of the above treat each chain separately, useful, but insufficient for cross-ledger flows. A concrete illustration is terrorist-financing related activity surrounding the Sri Lanka Easter attacks: an event-study on Bitcoin revealed abnormal volume through mixers and unregulated exchanges in the pre-event window; forward tracing then showed conversion to Ripple (XRP) and continued laundering on that ledger [54]. This case makes the cross-chain need explicit: without integrated address/entity linking and real-time exchange/bridge coverage, sophisticated actors can exploit siloed detectors.

While multi-chain comparative studies are valuable, they are insufficient for tracking sophisticated actors who exploit the seams between ecosystems [155], [156]. The critical open challenge is moving from parallel, side-by-side analysis to integrated, entity-centric detection that can follow illicit activity as it hops across chains, bridges, and exchanges.

D. CHALLENGES AND LIMITATIONS

Several critical and interrelated challenges permeate cryptoasset anomaly detection, spanning technical, behavioral, and regulatory dimensions.

First, the scarcity of accurately labeled data constitutes a fundamental obstacle. Confirmed illicit addresses are exceedingly rare relative to legitimate activity, resulting in severe class imbalance. This imbalance significantly impedes supervised learning, which depends on high-quality labeled datasets. The inherent pseudonymity of blockchain systems further complicates ground-truth validation. Consequently, researchers must explore semi-supervised, self-supervised, or active-learning strategies to leverage unlabeled data effectively and enhance model robustness in detecting anomalies. To alleviate these data constraints, researchers are encouraged to utilize and contribute to community-maintained repositories such as the GraphSense TagPacks [26] and other curated, publicly documented label sets. Promoting such open benchmarks, alongside rigorous reporting standards, is essential to address label scarcity and ensure reproducible validation across the field.

Second, scalability and real-time constraints remain pressing issues. Blockchain transaction volumes continuously grow, demanding highly efficient algorithms for anomaly detection capable of processing massive data flows at high velocity. Real-time detection at block-time granularity is essential to prevent financial losses and mitigate ongoing threats like smart contract exploits. Achieving timely, accurate anomaly detection with sub-second inference and manageable false-positive rates is computationally intensive, particularly for advanced methodologies like network analysis or complex machine-learning models. Therefore, further research into scalable, streaming anomaly-detection methods is crucial. A major practical challenge is the computational cost of advanced detection methods and its impact on real-time feasibility. Complex models like graph neural networks (GNNs) exemplify this issue. The time complexity of a single graph convolutional layer is often $O(|E|F' + |V|FF')$, where $|V|$ is the number of nodes, $|E|$ is the number of edges, and F/F' are the input/output feature dimensions. For a full model with multiple layers, this can scale to $O(Kmd + Knd^2)$ where K is the number of layers, m/n are edges (transactions) /nodes (addresses), and d is the feature dimension [157]. Given that blockchain transaction graphs can contain millions of nodes and hundreds of millions of edges, this cost can be prohibitive for real-time model retraining, which is a key reason why achieving sub-second inference at block-time granularity remains a significant challenge [158]. While inference is generally faster than training, latency can still bottleneck high-frequency scenarios. To mitigate this, many successful approaches employ subgraph sampling to confine computation to localized neighborhoods. For instance, the FraudLens framework [124], using graph restructuring, reported completing its experiment on the entire Elliptic dataset in under a minute on a powerful server. To make GNNs scalable for even larger graphs like Ethereum's full transaction history, many successful approaches employ subgraph sampling strategies. The HGATE framework [123], for example, avoids full-graph training by extracting smaller, localized "micro interaction subgraphs" around target accounts, which enables efficient mini-batch training while still capturing relevant behavioral patterns. This highlights a crucial trade-off: localized subgraph methods are computationally efficient and can fit on a single GPU, but they risk missing broader, collective anomalies that are only visible at a global scale. Full-graph analysis provides more comprehensive context but at a significant computational cost. Therefore, real-time deployment feasibility depends on striking a balance. Current research suggests a hybrid approach is most practical: using fast, subgraph-based methods like HGATE for real-time signal generation, while potentially running more comprehensive, full-graph analyses asynchronously to ensure network-wide coverage.

Third, distinguishing benign yet privacy-preserving behaviors from malicious obfuscation requires sophisticated behavioral modeling and nuanced feature engineering. Users increasingly adopt non-custodial wallets and other privacy-focused tools for legitimate reasons, such as ideological beliefs or data sovereignty concerns. However, criminals frequently exploit these tools due to their pseudonymous nature and absence of KYC procedures. Effectively addressing this ambiguity demands advanced analytical techniques that transcend basic transaction metrics and incorporate behavioral insights. Expanding on the challenge of behavioral ambiguity, the field needs a deeper integration of formal privacy-preserving analytics, where the very tools designed to protect user privacy can hinder anomaly detection. Blockchain users increasingly employ mixers, e.g. CoinJoin protocols or Tornado Cash, and privacy coins like Zcash which employ techniques like zero-knowledge proofs (ZKPs) that offer legitimate users enhanced confidentiality. This dual-use ambiguity forces detectors to distinguish benign privacy-enhanced behavior from malicious laundering. In practice, even privacy mechanisms leave telltale patterns. For example, mixing services often have characteristic input/output structures or timing signatures; simple heuristics exploiting these can identify over 92% of CoinJoin-style transactions despite their obfuscation [138]. Likewise, analyses of privacy-centric blockchains reveal trade-offs: Zcash's zero-knowledge shielded pool provides anonymity, yet repetitive usage patterns allowed clustering of 87.5% of addresses and linking a quarter of "anonymous" transactions to known entities (miners, founders), undermining its privacy in practice [80]. These examples highlight that privacy techniques can be partially pierced by analytical methods. Similarly, federated learning and secure multi-party computation (MPC) have been proposed to let exchanges or nodes jointly train anomaly detectors without sharing raw data, aligning with data protection regulations [159], [160]. Such approaches can preserve confidentiality (each party keeps its own dataset) but come with higher complexity and potential performance hits (e.g. communication overhead, convergence issues). Thus, privacy-preserving analytics in blockchain must balance detectability vs. privacy: stronger privacy tools (mixers, encrypted transactions) make it harder to spot illicit behavior, while privacy-preserving detection frameworks (differential privacy, federated models) safeguard user data at the cost of some sensitivity. Effective solutions will likely combine multiple techniques, for instance, incorporating privacy-aware heuristics into anomaly models, to ensure that legitimate privacy is upheld even as illicit abuse of privacy tools is aggressively detected.

Finally, the challenge of cross-chain anomaly detection is becoming increasingly pertinent. Attacks such as bridge exploits and flash-loan manipulations often leave traces distributed across multiple blockchain ecosystems, complicating detection due to fragmented and siloed data sources. Enhancing interoperability and developing detection methods capable of integrating multi-chain data streams are urgent areas for future research, necessary to effectively identify complex cross-chain anomalies.

E. FUTURE RESEARCH DIRECTIONS

Addressing these multifaceted challenges requires concerted research across technical, behavioral, and regulatory dimensions. Several promising research directions emerge clearly.

First, developing hybrid methodologies that integrate strengths from different detection categories represents a fertile area for future investigation. Graph Neural Networks (GNNs) combining network topology with machine learning classification exemplify such approaches, merging structural insights with data-driven detection capabilities. Similarly, rule-augmented machine learning pipelines that leverage heuristics to pre-select anomaly candidates for deeper analyses promise both interpretability and enhanced accuracy. Recent work also explores combining diverse mathematical anomaly indicators using AI techniques like Boltzmann machines to create more robust signals or integrating predictive AI with facilitation AI within organizational frameworks like DAOs [161], [162]. Formalizing design patterns and best practices for these hybrid systems could streamline development and improve reliability. This susceptibility highlights a critical operational challenge as attackers continuously evolve their strategies to bypass static filters, anomaly detection models inevitably suffer from drift and performance degradation. Consequently, deploying adaptive retraining pipelines and continuous drift detection mechanisms is as important as the initial model selection to mitigate these adversarial shifts.

Recently, emerging paradigms like Graph Foundation Models (GFMs) are opening new avenues in graph-based anomaly detection. GFMs represent a paradigm shift in graph machine learning. Reference [163] proposes a large-scale pre-training framework on heterogeneous transaction graphs. The results show that GFMs can be fine-tuned to various tasks, including anomaly detection, achieving strong accuracy with minimal supervision. The promise of emergent capabilities, e.g., in-context learning, zero-shot generation, and task homogenization across node, edge, and graph levels, could help unify the fragmented landscape of graph-based anomaly detection approaches. Building on this concept, GNN+LLM hybrid approach [164] fuses blockchain transaction graphs with cross-chain textual signals. By leveraging pre-trained language models alongside structural embeddings, they capture anomalies hidden both in graph topologies and semantic patterns, promising especially for detecting fraudulent behaviors embedded in multi-chain settings.

Second, advancing methods robust to label scarcity and severe data imbalance is critical. Techniques such as semi-supervised, self-supervised, and transfer learning can exploit unlabeled or partially labeled data, significantly improving anomaly detection in data-scarce environments. Furthermore, synthetic data generation approaches, designed to emulate diverse legitimate and illicit behaviors, could further alleviate data constraints and facilitate rigorous model evaluation.

Third, improving the interpretability of sophisticated machine learning models remains vital. Advanced ML and deep learning models often lack transparency despite their high accuracy. Research should prioritize Explainable AI (XAI) techniques tailored specifically for cryptoasset anomaly detection, employing attention mechanisms, saliency mapping, or post-hoc interpretation methods to elucidate the decision-making process of these powerful yet opaque models.

Fourth, developing scalable, real-time anomaly detection systems capable of processing high-throughput blockchain data streams is paramount. Techniques leveraging online learning, reinforcement learning, and hardware acceleration (e.g., GPUs, TPUs, distributed computing) warrant exploration to achieve timely, accurate detection at the scale and speed required by contemporary blockchain networks.

Finally, establishing standardized benchmarks and datasets is essential to enabling fair, consistent comparisons across methods. Creating labeled, timestamped datasets covering major cryptoassets and cross-chain interactions, accompanied by standardized evaluation metrics like precision-recall curves and time-to-detect metrics, would significantly advance methodological rigor and facilitate cross-study comparisons.

V. CONCLUSION

Growth has transformed the cryptoasset ecosystem into a major financial market involving substantial economic activity and a large Decentralized Finance (DeFi) sector. Correspondingly, the attack surface for fraud, market manipulation, and protocol-level exploits has expanded. Globally, regulatory frameworks are also maturing, imposing greater scrutiny and evolving compliance demands, which includes exploring new concepts of secondary liability [165].

This SoK has mapped 103 studies on cryptoasset anomaly detection across statistical analysis, network analysis, machine learning and heuristic-based methods. The comparative analysis reveals inherent trade-offs: statistical analysis offers interpretability but faces data distribution sensitivity, network analysis leverages topology effectively but struggles with scalability, machine learning provides powerful pattern recognition but often requires significant labeled data and can lack transparency. At the same time, heuristic-based methods excel with known threats via expert rules but fail against novel patterns and require ongoing updates. Across these approaches, persistent challenges hinder progress, notably the scarcity of labeled data and class imbalance, the computational demands of real-time detection at scale, the ambiguity between privacy techniques and malicious obfuscation, and the complexity of cross-chain activity analysis.

Addressing these significant challenges suggests several key directions for future research. Promising directions include developing hybrid methodologies like GNNs or rule-augmented ML, advancing techniques robust to label scarcity such as self-supervised learning and synthetic data generation, enhancing model interpretability through Explainable AI, creating highly scalable real-time systems, and crucially, establishing standardized benchmarks and datasets for rigorous comparison. Advancing cryptoasset anomaly detection is vital not merely as an academic exercise but as a crucial requirement for market integrity, user protection, and the responsible integration of digital assets into the global financial system, demanding robust, explainable, and adaptive solutions.

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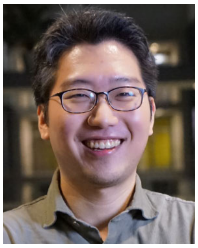

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